

BIOLOGY GRADE 11: EXCRETION AND HOMEOSTASIS IN ANIMALS

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 3.3 (7 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	11
Subject	Biology
Strand	Strand 3.0: Anatomy
Sub-Strand	Sub-Strand 3.3: Excretion and Homeostasis in Animals
Total Duration	7 lessons × 40 minutes = 280 minutes total
Sub-Strand Content	– Excretory structures and products in animals (insects, birds, freshwater and marine fish) – Internal structure of the mammalian kidney (cortex, medulla, pelvis, structure and function of the nephron) – Process of urine formation in mammals – Internal structure of the mammalian skin – Role of the liver and skin in excretion – Homeostatic regulation in humans (osmoregulation, thermoregulation, blood sugar regulation, including diabetes mellitus and diabetes insipidus) – Liver diseases and disorders (cirrhosis, hepatitis, fatty liver disease)
Learning Outcomes	By the end of the sub-strand, the learner should be able to: - Relate excretory structures to their products in animals (insects, birds, freshwater and marine fish). - Relate the internal structure of the mammalian kidney to its function. - Describe the process of urine formation in mammals. - Relate the structure of the mammalian liver and skin to their excretory functions. - Illustrate homeostatic regulation in humans. - Reflect on the importance of excretion and homeostasis in animals.
Core Competencies	– Critical Thinking and Problem Solving: The learner asks questions to be clear on what they need while discussing how animals in the locality respond to changes in the environment (Lesson 6: basking, panting, burrowing, hibernation, aestivation). – Digital Literacy: The learner uses digital technology effectively while using digital devices to watch animations on the process of urine formation in mammals (Lesson 3, with shared group viewing). – Communication and Collaboration: Learners work in peer groups to dissect, model, label and present the internal structures of the mammalian kidney (Lesson 2).
Core Values	– Social Justice: The learner shares resources equitably while modelling and labelling the internal structure of a mammalian kidney, including fair sharing of dissecting tools and modelling materials. – Responsibility: The learner engages in assigned roles and duties while dissecting a mammalian kidney to observe the internal structure.
Science & Engineering	– Asking Questions: Learners add questions to the Driving Question Board (DQB) after observing the dog, lizard, goat and fish,

Practices	and refine them as each lesson adds evidence. – Developing and Using Models: Learners draw and label a kidney, build a clay/chart nephron model, and build flowcharts, then revise one cumulative explanatory model each lesson. – Planning and Carrying Out Investigations: Learners dissect a fresh kidney from a local butchery and record observations safely. – Analyzing and Interpreting Data: Learners compare nitrogenous wastes across animals, trace fluid composition along the nephron, and interpret diabetes case-study data. – Constructing Explanations: Learners explain why animals differ in how they lose heat, water and wastes using structure-function evidence. – Engaging in Argument from Evidence: Learners challenge misconceptions, such as sweating being only for waste removal, using evidence. – Obtaining, Evaluating and Communicating Information: Learners use charts, animations, photomicrographs and a resource person, then present findings to peers.
Pertinent & Contemporary Issues (PCIs)	– Environmental Conservation: The learner observes laboratory safety while dissecting the mammalian kidney to observe internal structure, including proper biohazard disposal of specimens. – Waste Management: The learner appreciated the need to conserve biodiversity while observing and discussing how animals in the locality respond to changes in the environment. – Health Education: Learners explore liver disorders (cirrhosis, hepatitis, fatty liver disease) and metabolic disorders (diabetes mellitus, diabetes insipidus) and discuss prevention and management (Lessons 4 and 5).
Career Connections	– Veterinary Medicine: Understanding how livestock such as goats and dogs regulate water, heat and wastes. – Human Medicine and Nursing: Diagnosing and managing kidney disease, liver disease and diabetes. – Medical Pathology and Laboratory Technology: Examining tissue structure of kidney, liver and skin. – Physiological Research: Studying feedback mechanisms and adaptations to stress. – Environmental Conservation and Wildlife Management: Understanding how wild animals in Kenya's Rift Valley, Tsavo and Lake Victoria cope with heat, drought and changing conditions.
Focus for Lessons	This unit uses one everyday Kenyan puzzle, a panting dog, a motionless lizard, goats producing concentrated urine and freshwater fish producing dilute urine, to drive learning about excretion and homeostasis. Learners predict, gather evidence through charts, a kidney dissection, animations and case studies, and make sense of it through negative feedback ideas. They revise one cumulative model and track growing understanding on the DQB, using locally available materials and shared digital devices.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How do different animals maintain a balanced internal environment and eliminate toxic wastes despite drastic changes in their external conditions? KICD KEY INQUIRY QUESTIONS: 1. How do animals get rid of waste products? 2. How do animals cope with environmental changes?
Anchoring Phenomenon	On a hot afternoon in a local community, a dog pants rapidly with its tongue hanging out, while a lizard sits motionless on a warm rock. Nearby, goats drink water yet produce concentrated urine, whereas local freshwater fish continually excrete large amounts of dilute urine. All these animals face the same environment and the same need to get rid of wastes and keep the body stable, yet each behaves very differently. Students must ask why a hot lizard stays still on a hot rock, why a dog pants, why goats and fish handle water so differently, and how each animal keeps its internal conditions steady and removes toxic wastes.

Storyline Thread	Lesson 1: Insects, birds, freshwater fish and marine fish have specialized excretory structures adapted to eliminate specific nitrogenous wastes (uric acid, ammonia, urea) based on water availability. Learners explore charts and reference texts to compare excretory organs and products. Leads to: How do mammals (like humans) process and filter their internal liquid wastes? Lesson 2: The mammalian kidney contains distinct regions (cortex, medulla, pelvis) and functional units (nephrons) designed for blood filtration. Learners dissect a locally obtained kidney (from a butchery), draw the internal structures and build and label a model with peers, following safety and biohazard disposal procedures. Leads to: How exactly does the nephron turn filtered blood into urine? Lesson 3: Urine formation involves ultrafiltration in the glomerulus followed by selective reabsorption and secretion in the nephron tubule. Groups watch animations on shared devices and trace changes in fluid composition. Leads to: What other organs in mammals assist in excretion besides the kidney? Lesson 4: The skin and liver possess specialized structures that eliminate metabolic wastes and process toxic substances. Learners observe photomicrographs and illustrations and discuss deamination, bile production and sweat secretion, and hear from a resource person on liver disorders (cirrhosis, hepatitis, fatty liver disease). Leads to: How does the body keep internal water, temperature and sugar levels constant during external changes? Lesson 5: The body uses negative feedback control to regulate water levels (osmoregulation, ADH) and blood glucose (insulin and glucagon). Learners construct feedback flowcharts and evaluate case studies of diabetes mellitus and diabetes insipidus. Leads to: How does the body keep core temperature stable when the outside weather changes drastically? Lesson 6: Thermoregulation involves physiological mechanisms (sweating, vasodilation, vasoconstriction) and behavioral adjustments to control body heat. Learners analyze feedback loops and discuss local animal behaviors (basking, panting, burrowing, hibernation, aestivation), explaining the dog and lizard in the anchoring phenomenon. Leads to: Why are excretion and homeostasis so vital for the overall survival of living organisms? Lesson 7: Maintaining a constant internal environment (homeostasis) and clearing metabolic waste (excretion) are essential for cellular survival and overall health. Learners synthesize evidence into a final explanatory model, reflect on the importance of excretion and homeostasis, and present solutions to DQB questions. Unit reflection (no further question).
Supporting Phenomena	– A person who has run in the sun (such as a athlete training in Iten) sweats, turns red and feels thirsty, but passes little, dark urine. – Birds and insects excrete a whitish, semi-solid paste instead of liquid urine, while marine fish live in salty water yet do not dehydrate. – A person with untreated diabetes mellitus is constantly thirsty, passes a lot of urine and has sugar in the blood, which suggests that body regulation has failed. – Animals in Kenya's dry areas burrow, aestivate or hibernate during extreme weather, and survive by reducing activity and water loss.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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LESSON 1 (80 minutes (double lesson)): Same Hot Afternoon, Different Bodies: Launching the Excretion and Homeostasis Puzzle

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To launch the anchoring phenomenon (a panting dog, a still lizard, goats producing concentrated urine and freshwater fish producing dilute urine on the same hot afternoon) and help learners compare the excretory structures and nitrogenous products of insects, birds, freshwater fish and marine fish in relation to water availability.
Knowledge	Learners relate excretory structures and products in animals (insects, birds, freshwater and marine fish): ammonia (freshwater fish, via gills and dilute urine), urea (marine cartilaginous fish and mammals), uric acid (insects via Malpighian tubules, and birds via kidneys). Learners recognise that the form of nitrogenous waste depends on water availability.
Skills	Developing and using models (initial mental model of how an animal keeps its body stable and removes waste); asking questions (sticky notes for the DQB); analysing and interpreting data from charts and reference texts; constructing explanations; communicating with peers (Think-Pair-Share).
Attitudes	Core competency: Critical Thinking and Problem Solving, as learners ask questions to be clear on what they need while discussing how animals in the locality respond to changes in the environment. Value: Responsibility, as learners take on assigned group roles. PCI: Environmental Conservation, as learners appreciate the need to conserve biodiversity while observing and discussing how animals in the locality respond to environmental changes.
Key Inquiry Question	1. How do animals get rid of waste products? 2. How do animals cope with environmental changes?
Purpose in Storyline	This is the opening lesson. It surfaces learners' ideas about waste removal and body stability, sets up the Driving Question Board, and builds the comparative base (waste products linked to water availability across insects, birds and fish). It ends with the question: How do mammals (like humans) process and filter their internal liquid wastes? That leads to Lesson 2 on the mammalian kidney.
Safety Notes	This lesson uses no live animals or dissection. If learners observe animals outdoors, they must keep a safe distance and not touch dogs, lizards or livestock. Learners wash hands after handling shared charts and materials. Make sure no learner is left standing in direct sun during the outdoor-observation talk. No kidney dissection happens in this lesson; dissection safety (gloves, aprons, sharp-tool rules, handwashing) will be taught before the later dissection lesson.

B. LESSON OVERVIEW

Learners begin with the puzzle of a hot afternoon in a Kenyan community: a dog pants, a lizard sits still on a warm rock, goats drink water yet produce concentrated urine, and freshwater fish excrete large volumes of dilute urine. Learners first draw an initial model of how one animal (dog, lizard or human) handles excess heat or waste, and share their ideas in pairs. They then use charts and reference texts, working in groups of 4-6 with shared copies, to compare excretory organs and nitrogenous wastes (ammonia, urea, uric acid) in insects, birds, freshwater fish and marine fish, and connect these to water availability. In the explain phase they use the evidence to revise their thinking about the fish and goat part of the puzzle. The class then launches the Driving Question Board with sticky-note questions, and finally makes a first class model that shows what they know and what is still unexplained, leading to the next question: How do mammals (like humans) process and filter their internal liquid wastes?

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: excretion, homeostasis, animals, nitrogenous  READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms: excretion, homeostasis, animals, nitrogenous	Learners look at the phenomenon (teacher reads the scenario aloud and sketches the four animals on the board, or shows a poster): a dog panting, a lizard still on a hot rock, goats drinking yet producing concentrated urine, and freshwater fish excreting large amounts of dilute urine near Lake Victoria or a local dam. Individually (4 minutes), each learner draws an initial mental model/diagram showing how ONE animal (dog, lizard or human) handles excess heat or internal waste, using arrows and labels, and writes a one-sentence prediction: 'I think the dog pants because...'. Learners then Think-Pair-Share with a neighbour (3 minutes) and add or change one thing in their drawing using a different colour pen.	Read the scenario slowly, then say: 'Same hot afternoon, same village, same need to stay stable and get rid of wastes. Why do these animals behave so differently?' Give WAIT TIME of 10-15 seconds before taking any answer. Say: 'Draw what you think is happening INSIDE the animal. There are no wrong drawings today; I want your honest first thinking.' Circulate and look for ideas about sweat, water loss, heat, urine and wastes without correcting. Ask 2-3 learners at each desk cluster: 'What does your arrow show?' During the share, cold-call 4 pairs from different parts of the room (not volunteers only): 'Tell us what your partner added to the drawing.' Record recurring ideas on the board under 'Our first ideas' (e.g., 'panting cools the body', 'fish have too much water', 'lizard saves energy'). Do not confirm or correct. Collect drawings later and keep them in a folder for the Model Building phase and the end-of-unit comparison. For large classes, have back-row learners share with a pair behind them, then take answers from each row.	Think-Pair-Share, with an initial model drawing. Individual thinking first gives every learner time to put their own ideas on paper, and pair talk lets them test and refine ideas in a low-risk setting. The drawn initial model makes hidden preconceptions visible (e.g., 'waste is just sweat' or 'all animals urinate the same way') so they can be revisited and revised later in the storyline.	Observable indicators: (1) Learner's drawing includes at least one label for heat, water or waste and an arrow showing movement or loss. (2) Learner can state a prediction using 'I think... because...'. (3) Learner adds or changes at least one idea after the pair talk (different colour). Note common ideas and misconceptions (e.g., 'only mammals excrete', 'sweating is the only way to lose waste', 'fish do not urinate') on a quick tally sheet. Oral probe: 'What made you draw that arrow?'
Observe Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: excretion	In groups of 4-6 (with assigned roles: reader, recorder, chart-keeper, reporter), learners explore the charts and reference texts on excretory structures in insects, birds, freshwater fish and marine fish, one set per group. Using a comparison table (Animal Excretory organ/structure Main nitrogenous waste Water used to	Distribute charts/reference texts, one set per group, and display one chart at the front for the whole class. Say: 'You are scientists comparing evidence. Look for patterns between where an animal lives and the waste it makes.' Model the first row of the table on the board using the freshwater fish. Set a timer for 20 minutes.	Structured evidence comparison using a data table (a pattern-finding strategy). Organising the same categories across different animals makes patterns visible: less water available means a less toxic, less soluble waste (uric acid), while plenty of water means ammonia can be flushed away. Group roles and shared materials	Observable indicators: (1) Group table has all four animal groups filled in with organ and nitrogenous waste. (2) Learners correctly pair ammonia with freshwater fish, urea with marine cartilaginous fish, and uric acid with insects and birds. (3) Learners cite chart or text evidence when explaining. (4) Each learner can name their role

homeostasis animals nitrogenous  READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: excretion homeostasis animals nitrogenous	remove waste (a lot/some/very little) Habitat/water availability), they record evidence, e.g., freshwater fish (e.g., tilapia from Lake Victoria): gills and kidneys, ammonia, lots of dilute urine; marine fish: gills with chloride cells, kidneys, urea/trimethylamine oxide in cartilaginous fish, very little concentrated urine; insects (e.g., locust, housefly): Malpighian tubules, uric acid, almost solid paste; birds (e.g., chicken, ostrich, flamingo at Lake Nakuru): kidneys, uric acid, white paste in droppings. Groups then answer two guiding questions: 'Which animals need to save water, and what waste do they make?' and 'Which animals have plenty of water, and what waste do they make?' Groups share a few shared charts, rotating every 8 minutes if materials are limited.	Circulate with a checklist; ask: 'Which column shows how much water this animal can spare?' and 'What do you notice about the animals that make uric acid?' Give WAIT TIME of 10-15 seconds after each probe. Ensure every learner has a role and that the quietest learner reads or records at least once; cold-call 3 groups to report one row of their table. Prompt for evidence: 'Which chart or sentence tells you that?' Redirect misconceptions gently: 'Check the text again, what does it say about water use?' For groups finishing early, ask them to extend their table by adding the lizard and the goat from the phenomenon and to make a prediction about their waste.	support equitable participation in large classes.	and what they contributed. Teacher uses an oral probe at each group: 'Show me the line in the text that supports your column.' Record groups needing support.
Explain Phase  VIDEO: No closely matching video in the ARES library for this activity.  Search ARES for videos Search terms: excretion homeostasis animals nitrogenous  READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: excretion homeostasis animals	Learners return to individual thinking first: they write 2-3 sentences answering 'Why do freshwater fish excrete lots of dilute urine and ammonia, while insects and birds excrete a semi-solid uric acid paste?' using at least two pieces of evidence from their tables. They then Think-Pair-Share and join another pair to build a four-person explanation. Next the class discusses the phenomenon: 'What does this tell us about the goat and the freshwater fish?' Learners use their data to explain that fish living in water take in water by osmosis and must remove it as dilute urine, while terrestrial animals conserve water. They also name what is still NOT explained: how the goat produces concentrated urine, why the dog pants, and why the lizard	Say: 'Use your evidence. Complete this sentence: Animals that have little water make ____ because ____.' Give WAIT TIME of 10-15 seconds, then allow pairs to talk for 2 minutes. Cold-call 5 learners across the room to share, asking each: 'What evidence supports that?' and then 'Who can add to or challenge that idea?' Draw a simple line on the board from 'Plenty of water' (ammonia) to 'Very little water' (uric acid), and ask learners to place the animals on it. Link back: 'Now look at the goat and the fish from our hot afternoon. What can we say about each, and what can we NOT yet explain?' Record on a chart under 'What we can explain' and 'What we still cannot explain.' Highlight that the goat's concentrated urine and the dog's panting need	Claim-Evidence-Reasoning through Think-Pair-Share and a class discussion. Learners state a claim about waste type and water availability, support it with chart evidence, and reason about why. Naming what remains unexplained builds a productive 'need to know' that drives the next lessons and connects the evidence back to the full phenomenon.	Observable indicators: (1) Written explanation includes a claim, at least two pieces of evidence, and a link to water availability. (2) Learner uses the terms ammonia, urea and uric acid correctly. (3) Learner identifies at least one unexplained part of the phenomenon (goat urine, panting, lizard basking). (4) Learner revises the drawing with a 'Now I think...' note. Oral probe: 'Why would a bird not be able to make dilute ammonia urine like a tilapia?' Quick scan: 3-point check (claim, evidence, reasoning).

nitrogenous	sits on a hot rock. Learners add a 'Now I think...' box to their initial model drawing in a different colour.	something we have not yet studied. Avoid giving the mammalian kidney explanation; say: 'Hold that thought; it will be our next investigation.' Clarify vocabulary (excretion vs egestion; ammonia, urea, uric acid) briefly with a Kenyan example (e.g., white part in chicken droppings is uric acid).		
Driving Question Board (DQB) Creation  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: excretion homeostasis animals nitrogenous  READING: No closely matching reading in the ARES library for this activity. Search ARES for readings Search terms: excretion homeostasis animals nitrogenous	The class reads the Driving Question aloud: 'How do different animals maintain a balanced internal environment and eliminate toxic wastes despite drastic changes in their external conditions?' Individually, each learner writes 2 questions on sticky notes about the phenomenon (e.g., 'Why does a dog pant instead of sweat?', 'How do goats make concentrated urine?', 'Why does a lizard bask on a hot rock?', 'How do human kidneys clean blood?'). Learners place notes on the DQB, grouped under categories that emerge (Waste removal, Water balance, Temperature control, Body organs). Learners vote on 2-3 priority questions to investigate next, and a group volunteers to be 'DQB keepers' for the week.	Display the Driving Question on a large chart at the front. Say: 'Write questions you really want answered, questions that start with how or why.' Give WAIT TIME of 10-15 seconds before they begin writing. Model one good question and one weaker one ('What is urine?' vs 'Why does goat urine look darker than fish urine?'). Collect sticky notes row by row so that every learner contributes. Cold-call 4 learners to read and place their question, asking: 'Which part of the phenomenon does your question come from?' Cluster notes with the class: 'Which notes belong together? What should we call this group?' Connect to the KICD key inquiry questions: 'How do animals get rid of waste products? How do animals cope with environmental changes?' and place both on the board. Point out the question 'How do mammals (like humans) process and filter their internal liquid wastes?' (add it if no learner raises it) and link it to the goat. If wall space is limited, use a large manila sheet or a section of the chalkboard and photograph it. Assign DQB keepers.	Driving Question Board discussion. The DQB makes learners' curiosity visible and gives them ownership of the investigation sequence. Clustering questions helps them see relationships among excretion, osmoregulation and thermoregulation, and the board is revisited after each lesson to track growing understanding.	Observable indicators: (1) Each learner submits at least two questions. (2) At least some questions are testable or explanatory (how/why) rather than simple recall. (3) Learners can say which part of the phenomenon their question comes from. (4) Learners participate in sorting and voting. Teacher records the number of 'how/why' questions and notes learners who need support in question-writing. Oral probe: 'What would you need to find out to answer that?'
Model Building Phase	In their groups, learners build a first class/group model on a manila sheet or in their notebooks: a	Say: 'This is our first model. It will change as we learn more. Show what you know in one colour, and	Model Building and Revision. A first visible model, with unknown parts marked, captures current	Observable indicators: (1) Model shows all four animals with at least one label of heat, water or waste

VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: excretion, homeostasis, animals, nitrogenous READING: No closely matching reading in the ARES library for this activity. Search ARES for readings Search terms: excretion, homeostasis, animals, nitrogenous	labelled diagram showing the four animals from the phenomenon (dog, lizard, goat, freshwater fish) and, for each, what they think is happening to heat, water and waste inside the body. Using what they learned in this lesson, they add the nitrogenous waste and the excretory organs for fish, insects and birds, and use a different colour or question marks (?) for parts they cannot yet explain (goat's kidney, dog's panting, lizard's basking). Groups do a gallery walk (5 minutes) to compare models, leaving one 'I like/I wonder' note on another group's model. Finally, each learner completes an exit ticket.	mark what you do not yet know with question marks.' Give WAIT TIME of 10-15 seconds after setting the task. Provide one sheet of manila paper and markers per group; where materials are limited, allow notebook drawings. Circulate and ask: 'Where does the waste leave the body in this animal?' and 'What happens to water in this part of your model?' During the gallery walk, cold-call 3 learners to share one thing they noticed in another group's model. Close by linking to Lesson 2: 'We can explain fish, insects and birds, but the goat's urine and the human body leave us with a big question: How do mammals (like humans) process and filter their internal liquid wastes? Next lesson we will start finding out.' Hand out exit tickets: (1) One thing I now understand about nitrogenous wastes and water. (2) One question I still have. (3) Which animal's waste is hardest to explain and why? Collect models and keep them for revision in later lessons. Remind learners to bring any questions about kidneys and ask them to look at a goat or cow kidney at home or at a butchery (observation only, not handling).	understanding and shows gaps that need evidence. Gallery walk peer feedback encourages comparison and critique, and the model will be revised at the end of each lesson as the storyline develops.	for each. (2) Model correctly includes the excretory organ and waste for at least two of fish, insect, bird. (3) Model identifies gaps using question marks. (4) Peer feedback uses 'I like/I wonder' appropriately. Exit ticket: learner names one correct fact on nitrogenous wastes and water, and asks one relevant question. Tally exit tickets into 'secure', 'developing' and 'needs support' to plan Lesson 2.
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D. TEACHER REFLECTION

After the lesson, consider: Did learners' initial drawings reveal misconceptions (e.g., that all animals urinate in the same way, or that panting is just tiredness) that need targeted evidence later? Did every learner, including quieter ones in a large class, contribute a drawing, a role, and at least two DQB questions? Were charts and reference texts shared fairly among groups, and did rotation work? Which learners could link water availability to the type of nitrogenous waste, and who still needs support? Were the DQB questions mostly recall or genuine how/why questions, and how will you help learners improve them? Did the unexplained parts (goat urine, panting, lizard basking) create real curiosity about the mammalian kidney for Lesson 2? Note which Kenyan examples (tilapia from Lake Victoria, chickens, flamingos at Lake Nakuru, goats) were most engaging, and what you would adjust to manage time and materials next time.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We observed that freshwater fish, marine fish, insects and birds have different excretory structures (gills and kidneys, Malpighian tubules, kidneys) and make different nitrogenous wastes (ammonia, urea, uric acid), and that a dog pants, a lizard sits still on a hot rock, goats produce concentrated urine and freshwater fish produce large amounts of dilute urine.
What did I learn?	We learned that the type of nitrogenous waste an animal makes is linked to how much water is available to it: animals with plenty of water (freshwater fish) excrete toxic ammonia in dilute urine, while animals that must save water (insects and birds) excrete less toxic uric acid as a semi-solid paste, and marine fish and mammals make urea.
How does this explain the phenomenon?	We can explain that animals living in different environments have excretory structures adapted to remove nitrogenous wastes while conserving or removing water, but we cannot yet explain how a goat makes concentrated urine, why a dog pants, why a lizard basks, or how mammals such as humans filter their internal liquid wastes. That leads us to ask: How do mammals (like humans) process and filter their internal liquid wastes?

LESSON 2 (80 minutes (double lesson)): Inside the Goat Kidney: Cortex, Medulla, Pelvis and Nephrons

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To help learners discover, through dissection of a locally obtained goat or cow kidney, that the mammalian kidney has distinct regions (cortex, medulla, pelvis) and microscopic functional units (nephrons) designed for blood filtration, and to link this structure to how goats produce concentrated urine while freshwater fish produce dilute urine.
Knowledge	By the end of the lesson the learner should be able to: (b) relate the internal structure of the mammalian kidney to its function; identify the cortex, medulla, pelvis, ureter, renal artery and renal vein; state that the nephron is the functional unit of the kidney.
Skills	Planning and carrying out investigations (dissection), developing and using models (drawing and building a labelled clay/chart model), obtaining, evaluating and communicating information (presenting models to peers), asking questions and defining problems (DQB).
Attitudes	Responsibility: engages in assigned roles and duties while dissecting a mammalian kidney. Social Justice: shares resources equitably while modelling and labelling the kidney. Environmental Conservation and Waste Management: observes laboratory safety and disposes of biological waste properly. Critical Thinking and Problem Solving: asks questions to be clear on what they need.
Key Inquiry Question	How do animals get rid of waste products? How do animals cope with environmental changes?
Purpose in Storyline	Lesson 1 raised the puzzle of why the dog, lizard, goat and freshwater fish handle water and wastes so differently. Lesson 2 supplies the first structural evidence: the mammalian kidney is built for filtering blood, with regions and nephrons arranged for this job. It leaves learners asking: How exactly does the nephron turn filtered blood into urine? That question drives Lesson 3 (urine formation animation).
Safety Notes	Wear lab coats/aprons, disposable gloves and closed shoes; wash hands with soap and water before and after. Use scalpels only when the teacher permits, cutting away from the body on a tray. Only one trained cutter per group, others observe/record. Kidneys must be fresh, kept cool, and from a licensed butchery. No eating, drinking or touching the face. Dispose of tissue and gloves in a labelled biohazard bag/bin, to be buried in a deep pit or incinerated; disinfect trays and instruments with soapy water/bleach solution. Cover cuts. Keep sharp tools counted before and after.

B. LESSON OVERVIEW

Learners return to the hot-afternoon scene of the dog, lizard, goat and fish and predict what the inside of a goat's kidney must look like if it can make concentrated urine. In groups of 4-5 (about 10 groups for a class of 45-50) they dissect a goat or cow kidney from a local butchery, with defined roles, observe cortex, medulla and pelvis, and draw and annotate what they see. They explain how the structure suits blood filtration, then build and label a clay/chart model and peer-review other groups' models. The DQB is updated with new questions, especially about what the nephron does. The class model of how animals maintain a balanced internal environment is revised to include the kidney as the filtering organ.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners recall the phenomenon (panting dog, still lizard, goat)	Display a whole, uncut kidney in a tray (or a picture of one) and ask:	Predict-Observe-Explain with Think-Pair-Share. Making	Oral probes and sketch check. Look for: sketches showing at

 VIDEO: The kidney and nephron Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Kidneys (Flexbook) Source: CK-12  Search ARES for similar readings	producing concentrated urine, freshwater fish producing dilute urine). Individually, in 3 minutes, they sketch what they think the inside of a goat's kidney looks like and write one sentence on how it might help the goat make concentrated urine. They then Think-Pair-Share with a partner, compare sketches and write one prediction on what they expect to see when the kidney is cut open lengthwise (Predict-Observe-Explain, step 1). Time: 12 minutes.	"Yesterday we saw that the goat drinks water yet makes concentrated urine. If you could open this organ, what would you expect to see inside?" Give WAIT TIME of 10-15 seconds before anyone speaks, then 3 minutes of silent sketching. Circulate and ask 2 pairs: "What part of your sketch would let blood be cleaned?" Cold-call 4 learners from different groups to share predictions and record them on the board under 'Our Predictions'. Do not correct yet; say "Hold that idea, we will test it with evidence in a few minutes."	predictions first exposes learners' prior ideas (e.g., a kidney is hollow like a bean bag or just a bag of urine) so that the dissection evidence can confirm or challenge them, and pair talk lets every learner rehearse ideas in large classes.	least one inner region or tube; predictions mentioning blood supply or filtering; misconceptions such as 'urine is stored inside the kidney'. Tally on a class list how many learners predicted layers versus a hollow structure.
Observe Phase  VIDEO: The kidney and nephron - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES for similar videos  HTML: Kidneys (Flexbook) Source: CK-12  Search ARES for similar readings	In assigned roles (cutter, holder/observer, recorder, safety and disposal officer, materials manager) groups put on gloves and aprons and examine the external kidney: the bean shape, fatty capsule, renal artery, renal vein and ureter at the hilum. With the teacher's permission the cutter makes a longitudinal cut from the outer curve to the hilum. Learners observe and note: the darker outer cortex, the paler, striped triangular medulla (pyramids), the funnel-shaped hollow renal pelvis leading to the ureter, and the fine texture of the cortex. They use a hand lens to look for tiny granular dots (renal corpuscles) in the cortex. They draw a labelled, large, pencil drawing of the cut surface, then clean up and dispose of waste in the biohazard bin. Time: 30 minutes.	Before the dissection give a 3-minute safety briefing: "Gloves on, scalpel always cuts away from you, only the cutter touches the blade, and the safety officer counts the instruments." Demonstrate the first cut on one kidney at the front while groups watch. Ask groups to name roles aloud and rotate roles for the drawing so everyone handles tasks (Responsibility and Social Justice). Circulate with prompts: "Which region is darkest? Which is hollow? Where does the tube leave?" and "What does the colour difference tell you about blood supply?" Cold-call 3 groups to point to the cortex, medulla and pelvis on their specimen. Give WAIT TIME of 10-15 seconds after each question. Supervise disposal: all tissue, gloves and paper towels go into the biohazard bag, and trays are disinfected. With limited kits, let two groups share one dissection kit in turns while the other group drafts the diagram.	Predict-Observe-Explain (Observe). Direct examination of real tissue provides first-hand evidence that the kidney is organised into distinct regions and a collecting space; drawing forces learners to notice detail and decide what is important.	Annotated kidney diagrams plus observation checklist. Indicators: correct labels for cortex, medulla, pelvis, ureter, renal artery, renal vein; accurate relative position (cortex outside, medulla inside, pelvis central); title, scale or magnification note, and neat lines; safe handling observed (gloves on, cutting away from body, correct disposal). Record a tick or comment per group.
Explain Phase  VIDEO:	Groups compare their observations with the earlier	Ask: "Compare your sketch from earlier with what you saw. What	Claim-Evidence-Reasoning within Predict-Observe-Explain (Explain).	Oral probes and CER statements. Success indicators: learner uses

The kidney and nephron Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Kidneys and Homeostasis - Advanced (Flexbook) Related topic: not an exact match for this lesson. Source: CK-12 Search ARES for similar readings	predictions and discuss: "What did we get right or wrong?" Using a printed or board chart of a nephron (Bowman's capsule, glomerulus, tubule, loop of Henle, collecting duct) the teacher introduces the nephron as the microscopic unit that cannot be seen with the naked eye but is packed in the cortex and medulla. Groups write a claim-evidence-reasoning statement: 'The kidney is suited to filter blood because...' using evidence from their dissection (renal artery brings blood, the cortex contains many filtering units, medulla pyramids collect fluid, the pelvis funnels urine to the ureter). They connect it to the phenomenon, explaining why a goat's kidney, with a well-developed medulla, can produce concentrated urine while a freshwater fish's kidney mainly gets rid of excess water. Time: 18 minutes.	surprised you?" WAIT TIME of 10-15 seconds, then cold-call 4 learners. Introduce the nephron chart: "The cortex looked grainy. Those grains are tiny filters called renal corpuscles, part of each nephron. A goat kidney has about a million of them." Prompt reasoning: "If blood arrives through the renal artery, where must filtering happen, and where would the filtrate leave?" Ask groups to trace the path of fluid with a finger on the diagram from cortex to medulla to pelvis to ureter. Cold-call 3 groups to state the claim and evidence. Correct misconceptions explicitly, e.g., "The pelvis does not make urine; it collects it." Pose the bridging question: "Tissue structure shows where filtering happens, but how is urine actually made?"	Returning to predictions helps learners revise ideas using evidence and connect structure to function, while the nephron chart links what they saw at the visible scale to the microscopic units.	specific dissection evidence (e.g., 'the cortex is dark and granular, showing a rich blood supply') to support a function claim; correctly links pelvis to ureter and urine collection; identifies the nephron as the functional unit. Quick 3-question concept check quiz (1. Name the outer region. 2. Which region collects urine? 3. What is the functional unit?) with a show of fingers or mini-whiteboards.
Driving Question Board (DQB) Creation  VIDEO: The kidney and nephron - dubbed(KY) Source: Khan Academy (Kiswahili) Search ARES for similar videos  EXERCISE: Kidneys Practice Related topic: not an exact match for this lesson. Source: CK-12 Search ARES for similar readings	Each group writes 2 new questions on sticky notes or paper strips about what they still wonder (for example: How do the nephrons filter blood? Why is the medulla striped? Why does a freshwater fish not need a large medulla? What happens to useful substances like glucose from ugali? How does water loss in panting change urine?). They place the notes on the class DQB under categories (Structure, Process, Different animals, Homeostasis). They cluster similar questions, and mark which earlier questions have now been answered (e.g., 'Which organ filters blood in a goat?'). Time: 8 minutes.	Say: "What new questions has the kidney raised for you?" Allow WAIT TIME of 10-15 seconds, then ask each group to submit 2 questions. Cold-call 3 groups to read their questions aloud and place them on the board. Highlight one question: "How exactly does the nephron turn filtered blood into urine?" and say "This is the question we will investigate next lesson using an animation." Put a green tick beside questions answered and ask for the evidence that supports each tick. Photograph the DQB so it can be recreated if the wall space is needed.	Driving Question Board discussion. Learner-generated questions give ownership, show how structure evidence leads to new process questions, and set up a purposeful transition to Lesson 3.	Exit ticket and DQB quality. Indicators: questions that go beyond naming parts (ask how or why); links to the anchoring phenomenon (dog, lizard, goat, fish); at least one group asking about nephron function. Exit ticket: 'One thing I can now explain about the kidney is ____; one thing I still wonder is ____.'

Model Building Phase  VIDEO: The kidney and nephron Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Renin Production In The Kidneys <i>Related topic: not an exact match for this lesson.</i> Source: khan_academy  Search ARES for similar readings	Groups use modelling clay, cardboard, manila paper, coloured pencils and string (as vessels) to build and label a 3D or 2D longitudinal section of the kidney showing cortex, medulla (pyramids), pelvis, ureter, renal artery and renal vein, with an inset sketch of a nephron placed in the cortex and medulla. Using a different colour, they add to the class model of homeostasis a box labelled 'Kidney: filters blood and removes nitrogenous waste (urea) and regulates water', linking it to the goat, with a question mark for how the fish differs. Groups conduct a gallery walk: two members stay as presenters while others visit, use a peer-review checklist, and give one 'star' and one 'step' of feedback. Models are revised based on feedback. Time: 12 minutes.	Say: "Your model must show where blood enters, where filtering happens, and where urine leaves. Label every part." Ensure equitable sharing of materials (materials manager distributes equally; all members contribute). Circulate asking: "Which part of your model is the filter? How does this relate to the goat's concentrated urine?" WAIT TIME of 10-15 seconds after each question. For the gallery walk, rotate presenters after 3 minutes and cold-call 3 visiting learners to share a strength they saw. Collect the models and diagrams for display. Remind groups to wash hands, clean up and dispose of materials safely. Preview the next lesson: "We will watch an animation in groups of 5 around one device to see how a nephron makes urine."	Model Building & Revision with peer review. Building and labelling a physical model consolidates the structure-function relationship, and revising it after feedback shows that models are improved with evidence; adding the kidney to the class homeostasis model keeps the phenomenon central.	Peer review of physical models and teacher checklist. Indicators: all six structures correctly labelled; correct relative positions; nephron inset located in cortex/medulla; presenter can explain one structure-function link in a sentence; feedback is specific. Collect annotated diagrams for marking: 4 = all labels correct with function notes; 3 = labels correct; 2 = some errors; 1 = major errors.
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D. TEACHER REFLECTION

Did every learner, not just the cutter, handle the specimen or contribute to the drawing? Which predictions (hollow bag, layers, tubes) were most common and how did the evidence shift them? Were safety and disposal procedures followed without reminders? Which groups confused the pelvis with the medulla or imagined urine being made in the pelvis, and how will I address this at the start of Lesson 3? Were the large-class constraints (shared kits, time) managed without sacrificing quality? Did the DQB questions show curiosity about the nephron and the animals in the opening phenomenon?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed a goat/cow kidney with a dark outer cortex, a paler striped medulla, a hollow renal pelvis leading to the ureter, and blood vessels (renal artery and vein) entering and leaving at the hilum. Granular dots in the cortex suggested tiny filtering units.
What did I learn?	The mammalian kidney is organised into regions with different roles: the cortex contains many nephrons where blood is filtered, the medulla contains parts of nephrons and collecting ducts that help concentrate urine, and the pelvis collects urine and passes it to the ureter. The nephron is the functional unit of the kidney.
How does this explain the phenomenon?	Because the kidney receives a large blood supply and contains millions of nephrons arranged in the cortex and medulla, it can filter blood, remove urea and regulate water. This helps explain how a goat can drink water yet produce concentrated urine, while we still need to find out how nephrons turn filtered blood into urine and why freshwater fish produce dilute urine.

LESSON 3 (80 minutes (double lesson)): Following the Fluid: How the Nephron Filters Blood and Makes Urine

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use digital animations to trace how fluid composition changes along the nephron, so they can explain how a goat produces concentrated urine while a freshwater fish produces large volumes of dilute urine.
Knowledge	Describe the process of urine formation in mammals: ultrafiltration in the glomerulus, selective reabsorption in the proximal tubule, loop of Henle and distal tubule, secretion, and water reabsorption in the collecting duct under ADH control. Relate the structure of the nephron to each stage.
Skills	Developing and using models (tracing and annotating a nephron flowchart); analysing and interpreting data (comparing fluid composition in blood, filtrate and urine); constructing explanations; using digital devices to gather evidence; asking questions.
Attitudes	Responsibility in group roles when sharing a device; Social Justice by sharing devices equitably; curiosity and appreciation of how animals in the locality keep their bodies stable; Digital Literacy through careful use of animations.
Key Inquiry Question	1. How do animals get rid of waste products? 2. How do animals cope with environmental changes?
Purpose in Storyline	Lessons 1-2 established that excretory structures differ across animals and that the mammalian kidney has a cortex, medulla and pelvis containing nephrons. Lesson 3 explains HOW the nephron makes urine, giving the mechanism behind the goat's concentrated urine. It also raises the gap that the kidney is not the only excretory organ, which leads to Lesson 4 on other organs (liver, skin, lungs).
Safety Notes	Keep devices on stable desks away from water and any kidney specimens; wash hands if kidneys from butcheries were handled earlier in the week; devices are shared, so the Device Manager handles them and cables are kept off walkways; limit screen brightness and sit so all group members can see without crowding.

B. LESSON OVERVIEW

Students return to the opening scene: the goat drinks water yet produces concentrated urine, while the freshwater fish continually excretes dilute urine. In groups of 6-8 around one shared device, they predict what the kidney does to blood, watch animations of the nephron, and complete a 'Follow the Fluid' table tracing glucose, protein, urea and water in blood, filtrate and urine. They explain why glucose is absent from urine but urea is concentrated, then add the nephron process to the class model and the DQB, ending with the question: what other organs in mammals help with excretion besides the kidney?

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: The kidney and nephron Source: Khan Academy (English)	Students recall the goat and fish from the phenomenon and the kidney structure from Lesson 2. Individually they draw a quick sketch or write 2-3 sentences predicting: 'Blood enters the kidney'	Display the phenomenon picture or sketch (goat, fish, dog, lizard) on the board. Say: 'Yesterday we saw that the kidney has millions of nephrons. Today we ask: what exactly happens inside one	Think-Pair-Share surfaces prior ideas and misconceptions (for example, that urine is simply filtered blood, or that the kidney removes only waste). Individual thinking time ensures every	Look for: (1) at least 80% of learners have written a prediction before pairing; (2) number of pairs that mention both 'filtering' and 'taking back' of useful substances; (3) misconceptions recorded on

- US curriculum) Search ARES for similar videos READING: No closely matching reading in the ARES library for this activity. Search ARES for readings Search terms: excretion homeostasis animals ultrafiltration	carrying glucose, proteins, urea and water. What do you think comes out as urine, and what happens to the rest?' They then Think-Pair-Share with a neighbour and each pair records one prediction on a sticky note or in the exercise book: 'Urine contains ____ but not ____ because ____.' Groups of 6-8 are formed around the shared device with roles: Device Manager, Recorder, Reporter, Time-keeper, Skeptic (asks 'What is the evidence?') and Materials Monitor.	nephron to turn blood into urine?' Give 2 minutes of silent writing, then WAIT TIME 10-15 seconds before calling on anyone. Cold-call 4 pairs with: 'What do you predict leaves the blood and what stays?' Record all predictions on the board without judging; circle disagreements, for example whether glucose is found in urine. Say: 'Some of you think glucose is lost in urine. Keep that as a claim we will test with evidence.' Assign roles quickly so each learner has a duty, and rotate who handles the device.	learner commits to a prediction, creating a reason to watch the animation closely. This is the Predict step of Predict-Observe-Explain, so the later evidence is compared to their own ideas.	the board (e.g., 'glucose leaves in urine', 'all water leaves'). Oral probe: 'Why would the body not want to lose all the glucose from the ugali you ate?' Tally predictions to compare with post-animation understanding.
Observe Phase VIDEO: The kidney and nephron - dubbed(KY) Source: Khan Academy (Kiswahili) Search ARES for similar videos READING: No closely matching reading in the ARES library for this activity. Search ARES for readings Search terms: excretion homeostasis animals ultrafiltration	Groups watch the nephron animation clips on one shared device (a phone, tablet or laptop; clips are pre-loaded offline). Clip 1: blood flow into the glomerulus and ultrafiltration into Bowman's capsule under high pressure. Clip 2: selective reabsorption in the proximal convoluted tubule (glucose, amino acids, salts, water), the loop of Henle, secretion in the distal tubule, and ADH-controlled water reabsorption in the collecting duct. The group watches the first time without writing, the second time pausing at each region, and the third time to check details. Using the 'Follow the Fluid' table, they record for each region (renal artery blood, glomerular filtrate, after proximal tubule, after loop of Henle, urine) what is present or changed for glucose, proteins, urea, water and salts, using arrows or approximate amounts (high, medium, low, none). Each learner also annotates a blank nephron diagram with labels: glomerulus, Bowman's capsule, proximal tubule, loop of Henle, distal tubule,	Before starting, say: 'Your job is to follow one drop of fluid, like a matatu following its route, and write down what is added or removed at each stop.' Model pausing the clip at the glomerulus and ask: 'Why are blood cells and large proteins not in the filtrate?' WAIT TIME 10-15 seconds, then cold-call 3 groups. Circulate every 3-4 minutes: check that the Device Manager is pausing clips at the right points, the Recorder fills the table, and everyone contributes. Ask probing questions: 'What does the animation show happening to glucose in the proximal tubule? Where is the evidence?' 'Why is the urine more concentrated at the end?' Intervene where groups confuse filtration with reabsorption by asking: 'Which one is pressure-driven, and which one needs carrier proteins and energy?' For large classes, stagger sets of questions so two or three groups at a time get your attention, and use group Reporters to share. If a device fails, merge groups or project one clip for the class and keep the table activity going.	Predict-Observe-Explain: students compare observed animation evidence with their predictions. Tracing the fluid composition as a table and annotated diagram is a structured data-tracking approach that helps learners see patterns: what is filtered by size, what is reabsorbed by need, and what is concentrated. Using the 'follow the drop' analogy builds mental models of cause and effect along the nephron.	Check annotated nephron diagrams and the 'Follow the Fluid' table against indicators: (1) glomerulus labelled with ultrafiltration of small molecules (water, glucose, urea, salts) and exclusion of blood cells and proteins; (2) glucose shown as 100% reabsorbed by the end of the proximal tubule; (3) urea shown as concentrated in urine; (4) water reabsorption linked to the loop of Henle and collecting duct. Oral probes at each group: 'Which part of the nephron does filtering?' Mark each group on a quick checklist (secure / partially / not yet). Look for the Recorder and Skeptic actively using evidence language such as 'The animation shows...'.

	collecting duct, and writes what happens at each site. Groups without direct viewing access (back of the group) rotate seats at the half-way point so everyone sees the screen closely.			
Explain Phase  VIDEO: The kidney and nephron Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: excretion homeostasis animals ultrafiltration	Using their table and annotated diagrams, groups build a concept map or flowchart: Blood in renal artery → glomerulus (ultrafiltration) → Bowman's capsule → proximal tubule (reabsorption of glucose, amino acids, most water and salts) → loop of Henle (concentrating the medulla) → distal tubule (secretion, adjusting salts and pH) → collecting duct (ADH-controlled water reabsorption) → ureter → bladder → urine. They then write a claim-evidence-reasoning explanation for the phenomenon: 'The goat produces concentrated urine because ____ (evidence from nephron). A freshwater fish produces dilute urine because ____.' They link ADH to the dog panting on a hot afternoon: when the body loses water, more ADH is released, so the collecting duct reabsorbs more water. They also discuss what would happen if the proximal tubule failed to reabsorb glucose, linking to diabetes mellitus (glucose in urine) as a Kenyan health example. Groups then present a one-minute explanation to another group.	Say: 'Turn your table into a story: what is the journey of a glucose molecule and what is the journey of a urea molecule?' Give 8 minutes for the flowchart. Pose: 'The goat drinks water yet produces concentrated urine. Use your nephron evidence to explain.' WAIT TIME 10-15 seconds, then cold-call 3 different learners to explain one step each, building a chain. Challenge with: 'What would urine look like if the glomerulus was damaged and let proteins through?' Introduce ADH briefly: 'When the goat has lost water through sweating or has not drunk, the brain releases ADH. What does ADH do in the collecting duct?' Prompt connecting to freshwater fish: 'Fish take in water by osmosis. What would their kidneys need to do differently?' Correct misconceptions: urine is not simply filtered blood; reabsorption returns useful substances. Use peer review: groups swap flowcharts and add one 'star' (a correct feature) and one 'wish' (something missing). Validate correct language such as ultrafiltration and selective reabsorption.	Concept Mapping/Flowcharts convert the animation into a structured causal chain, making relationships between structure and function explicit. Claim-Evidence-Reasoning links evidence in the table to the phenomenon, and peer review forces learners to justify their reasoning. Connecting ADH to a negative feedback loop prepares the ground for the homeostatic feedback loops in later lessons.	Collect flowcharts and assess against indicators: (1) correct ordering of at least six stages; (2) ultrafiltration described as pressure-driven and size-dependent; (3) selective reabsorption described as returning glucose, amino acids, and water; (4) concentrated urine explained by water reabsorption controlled by ADH. Concept check quiz (5 questions, 4 minutes): (1) Name the site of ultrafiltration. (2) Why is glucose absent from urine in a healthy person? (3) Which hormone controls water reabsorption? (4) Name one substance secreted into the tubule. (5) Why does a goat's urine become more concentrated on a hot day? Score by show of fingers or quick tally; follow up where fewer than 70% are correct.
Driving Question Board (DQB) Creation  VIDEO: The kidney and nephron -	Groups return to the class Driving Question Board. They (1) place a sticky note with one answered question from earlier lessons, for example 'How does the kidney make urine?' with a short answer, and (2) write one new question	Say: 'Look at our board. Which questions can we now answer using nephron evidence? Move those to the Answered column and write the evidence.' WAIT TIME 10-15 seconds, then cold-call 3 groups to share. Gently guide	Driving Question Board discussions make learning visible and student-owned. Moving questions to 'Answered' with evidence shows progress, while new questions generate purpose for the next lesson. Sorting	Count the number of new questions per group and classify them as factual or deep (explaining why/how). Indicators of progress: at least 3 questions about other excretory organs; questions that link structure to

dubbed(KY) Source: Khan Academy (Kiswahili) Search ARES for similar videos READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms: excretion, homeostasis, animals, ultrafiltration	that the nephron evidence raises, for example 'What happens when the kidney fails?', 'Can the dog pant to remove urea?', 'Do fish nephrons have a loop of Henle?', or 'What other organs help remove waste?'. They sort questions into categories: kidney function, other organs, environmental responses, and disease. Each group reads out one new question.	towards the next lesson: 'We have focused on the kidney. But the dog is also panting and the lizard is basking. Is the kidney the only organ removing wastes?' Highlight any student question about sweating, breathing out carbon dioxide or the liver, and say: 'This question will drive our next investigation.' Ensure each group posts at least one question and that quieter learners get to read theirs out. Redirect off-topic questions politely by parking them on a 'Later' section.	questions helps learners see connections between the kidney, other organs and animal adaptations.	function; learners using terms such as 'reabsorption' and 'filtrate' correctly. Oral probe: 'How does your new question relate to the goat and the fish?'. Exit ticket (below) is given at the end of this phase or the next.
Model Building Phase VIDEO: The kidney and nephron Source: Khan Academy (English - US curriculum) Search ARES for similar videos READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms: excretion, homeostasis, animals, ultrafiltration	Individually and then in groups, learners revise the class model of how animals keep a balanced internal environment and remove wastes. Starting from their previous model (kidney structure and excretory structures of different animals), they add a labelled nephron diagram and arrows showing inputs (blood with glucose, urea, water, salts), processes (ultrafiltration, reabsorption, secretion, ADH control) and outputs (urine and reabsorbed substances returning to blood). They add a short label connecting the model to the goat (concentrated urine when water is scarce) and the freshwater fish (large volumes of dilute urine when water floods in). They mark a '?' on the model where they do not yet know how the body gets rid of other wastes (sweat, bile, carbon dioxide). Learners complete an exit ticket: (1) One thing I figured out about urine formation: ____ (2) One thing I am still wondering: ____ (3) Complete: Glucose is not in	Say: 'Open your model from the last lesson. What new parts must you add from the nephron evidence? What must you change?' Give 8 minutes for revision. Circulate and look for models that show a process rather than just labels. Pause and ask: 'Where on your model is the pressure that pushes fluid out of the blood? Where is the energy-using step?' WAIT TIME 10-15 seconds, then cold-call 3 learners to explain revisions. Use peer review: groups swap and give one comment, 'One thing I like is ____; one thing to add is ____.' Point to the '?' marks and say: 'Our model explains how the kidney removes urea and balances water. But the dog is panting and the goat is sweating. Which other organs might help? We will investigate this in the next lesson.' Collect exit tickets at the door and use them to plan reteaching.	Model Building & Revision requires learners to integrate new evidence into a cumulative explanation of the phenomenon, making their thinking visible and revisable. Marking gaps with '?' builds a need for further evidence. Peer review strengthens scientific argumentation and communication.	Assess the revised model using indicators: (1) nephron stages in the correct sequence with labels; (2) arrows show reabsorption returning to blood; (3) clear link between ADH and urine concentration; (4) goat and fish contrasted in the model; (5) at least one '?' identifying an unanswered question. Exit ticket sorting: green (can explain ultrafiltration and reabsorption), yellow (can name stages but not explain), red (confuses stages). Plan a 5-minute reteach at the start of the next lesson for red group learners.

	urine because ____.			
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D. TEACHER REFLECTION

After the lesson, reflect: Did every learner in the large groups get a chance to see the screen and contribute, or did one or two dominate the device? Which part of the nephron was hardest to understand (probably selective reabsorption versus secretion or the loop of Henle)? Did learners link the nephron process back to the goat and the fish without prompting? Which misconceptions persisted (for example, that urine is just filtered blood or that glucose is lost in urine)? Review exit tickets to decide who needs reteaching, and consider whether the animation pace or group size should change. Think about how to connect the next lesson on other excretory organs (liver, skin, lungs) to the questions learners posted on the DQB, and whether role rotation improved participation.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We watched animations of the nephron and traced how blood, filtrate and urine differ in glucose, protein, urea and water. We saw that blood cells and proteins stay in the blood, glucose is fully reabsorbed in the proximal tubule, and urine is concentrated by water reabsorption in the loop of Henle and collecting duct.
What did I learn?	Urine formation involves ultrafiltration in the glomerulus (small molecules forced into Bowman's capsule under high pressure), followed by selective reabsorption of useful substances and secretion of extra wastes and ions in the nephron tubule. ADH controls how much water is reabsorbed, so urine can be concentrated or dilute.
How does this explain the phenomenon?	A goat that loses water on a hot day releases more ADH, so its collecting ducts reabsorb more water and it produces small amounts of concentrated urine. A freshwater fish gains water by osmosis, so it produces large amounts of dilute urine. New question: What other organs in mammals assist in excretion besides the kidney?

LESSON 4 (80 minutes (double lesson)): Liver and Skin: Hidden Waste Processors Behind the Hot Afternoon

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners relate the structure of the mammalian liver and skin to their excretory and detoxifying functions, and use this evidence to explain how a panting dog, a goat and other animals remove metabolic wastes and keep the body stable on a hot afternoon.
Knowledge	By the end of the lesson, the learner should be able to: (d) relate the structure of the mammalian liver and skin to their excretory functions. The learner identifies hepatocytes, sinusoids, bile canaliculi, the epidermis, dermis, sweat glands and blood capillaries on photomicrographs and illustrations. The learner describes deamination (amino acids to urea), bile pigment production (breakdown of haemoglobin to bilirubin) and sweat secretion (water, salts and a little urea), and outlines the effects of cirrhosis, hepatitis and fatty liver disease.
Skills	Observing and interpreting photomicrographs (NGSS SEP: Analyzing and Interpreting Data); drawing and annotating structures (SEP: Developing and Using Models); constructing explanations from evidence (SEP: Constructing Explanations); asking questions of a resource person (SEP: Asking Questions); Critical Thinking and Problem Solving: asking questions to be clear on what they need.
Attitudes	Responsibility: engaging in assigned group roles and duties. Social Justice: sharing the limited charts, slides and devices equitably. Waste Management and Environmental Conservation: appreciating how organisms manage their wastes and how healthy living protects the liver.
Key Inquiry Question	1. How do animals get rid of waste products? 2. How do animals cope with environmental changes?
Purpose in Storyline	Lessons 1 to 3 gathered evidence on excretory products and organs in insects, birds and fish and on the kidney and nephron. Lesson 4 adds two organs not yet considered: the liver, which makes the urea the kidney removes, and the skin, which excretes sweat and cools the body. It closes a gap in the model: the kidney is not the only organ of excretion and homeostasis. It also leads to Lesson 5: How does the body keep internal water, temperature and sugar levels constant during external changes?
Safety Notes	Handle glass slides and microscopes carefully and report breakages immediately. Do not touch broken glass. Wash hands after handling any materials. If fresh kidneys from earlier lessons are still in the room, keep them covered in trays and wear gloves. Be sensitive to learners with family members who have liver disease or alcohol-related illness; keep discussion factual and respectful. The resource person must not give personal medical advice. Remind learners to avoid self-medication and to seek clinic advice.

B. LESSON OVERVIEW

Learners return to the hot-afternoon phenomenon (panting dog, still lizard, goats, freshwater fish) and ask: where does the urea in a goat's urine come from, and what does sweat do for a dog-like mammal such as a human in Kisumu at noon? They predict the roles of the liver and skin, examine photomicrographs, charts and illustrations of liver and skin histology in groups, discuss deamination, bile production and sweat secretion, and hear a resource person (a clinical officer, nurse or laboratory technologist) explain cirrhosis, hepatitis and fatty liver disease. They build explanations linking structure to function, update the Driving Question Board with the new question about keeping water, temperature and sugar constant, and revise the class model so that the liver and skin sit alongside the kidney.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
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Predict Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: excretion homeostasis animals liver  READING: No closely matching reading in the ARES library for this activity. Search ARES for readings Search terms: excretion homeostasis animals liver	(10 min) Learners look at the phenomenon image again (dog panting, lizard on rock, goats, freshwater fish). Individually, in their notebooks, they answer: (1) Goat urine contains urea. Where in the body is urea made? (2) Humans at a Mombasa beach sweat heavily. What is in sweat and why is it released? (3) What do you think the liver does besides storing food? They draw a quick sketch of where they think the liver and skin fit into the body's waste removal and write one confident idea and one uncertainty. Then Think-Pair-Share: they compare predictions with a partner and note one agreement and one disagreement.	Display the phenomenon image. Say: 'Last lesson we followed urea through the nephron. Today we ask: where did that urea come from, and what else is leaving the body on a hot afternoon?' Give 3 minutes of silent writing. WAIT TIME: 10-15 seconds after each question before taking any response. Say: 'I am not looking for the right answer yet; I want your best thinking.' Circulate and note common ideas (for example 'the liver only stores food', 'sweat is only water', 'urea is made in the kidney'). Cold-call 4 pairs: 'Pair 3, what did you agree on? Pair 7, what did you disagree on?' Record the predictions on the board under the heading 'Our first ideas' without correcting them. Ask one probing question: 'What would make you change your mind?'	Think-Pair-Share inside a Predict-Observe-Explain cycle. Making predictions in writing first surfaces preconceptions (for example that urea is made in the kidney), creates a need to know and gives learners a personal stake in the evidence they will see. Pairing lets learners rehearse ideas safely before sharing in a large class.	Observable indicators: at least 4 of 5 sampled notebooks show a prediction for each of the 3 questions; learners can state where they think urea is made; the teacher tallies on the board how many pairs think the kidney makes urea versus the liver. Oral probe: 'Why do you think that?' requires a reason, not only an answer. Misconceptions recorded for comparison in the Explain Phase.
Observe Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: excretion homeostasis animals liver  READING: No closely matching reading in the ARES library for this activity. Search ARES for readings Search terms: excretion homeostasis animals liver	(25 min) In groups of 5-6 (roles: materials manager, recorder, drawer, reporter, timekeeper) learners rotate through three stations with shared resources. Station A, Liver histology: examine a photomicrograph/chart of liver tissue (and a microscope slide where available). Identify hepatocytes, sinusoids, central vein, hepatic portal vein and bile canaliculi. Draw and annotate a labelled sketch and answer: how does blood flow past the hepatocytes? Station B, Skin histology: examine a skin section chart/photomicrograph. Identify epidermis, dermis, sweat gland, sweat duct, pore, blood capillaries, hair follicle and erector muscle. Annotate where sweat is made and released and how blood vessels near the surface help cooling. Station C, Function cards:	Set up stations before class using the available charts, photomicrographs and slides. Where there are not enough slides or charts, laminate or copy pages from the textbook so every group has something. Say: 'You are scientists reading tissue, not just labelling a picture. For each structure, ask: what is it, and what job does its shape suggest?' Assign roles and say: 'Materials managers, you are responsible for fair sharing; nobody keeps the chart for more than 8 minutes.' Ring a bell for rotation. Circulate with oral probes: 'Why are the sinusoids so wide and thin-walled? What does that allow between blood and hepatocytes?' and 'Why is the sweat gland coiled and surrounded by capillaries?' WAIT TIME: 10-15 seconds after each probe before prompting. Cold-call	Predict-Observe-Explain (observation step) combined with Analyzing and Interpreting Data. Rotating stations let a large class gather the same evidence with limited resources. The 'shape suggests job' prompt trains learners to reason from structure to function, the core idea of this sub-strand, and the function cards link the images to the chemistry of wastes so learners connect the liver and skin to the phenomenon.	Observable indicators: annotated sketches show at least 4 correctly labelled structures for the liver and 5 for the skin; the completed table has a correct entry for deamination (urea to the kidney), bile (bilirubin to the intestine) and sweat (water, salts, urea via the skin); learners can say why sinusoids and sweat gland capillaries are important. Oral probes: 'What would happen to urea in the blood if the liver could not deaminate amino acids?' Teacher checks and initials annotated diagrams during rotation.

	read short cards on deamination (excess amino acids lose the amino group which forms ammonia, then urea, carried by blood to the kidneys), bile (bilirubin from old red blood cells is excreted in bile, which also emulsifies fats) and sweat (water, salts and a little urea lost from the skin; evaporation cools the body). Complete a table: Structure, Substance removed or processed, Where it goes. Learners apply it to the phenomenon: a goat on a hot afternoon digesting dry grass, and a human walking in Nairobi in the heat. Every group takes about 8 minutes per station.	3 groups for each station at the end: 'Group 2, tell us one structure and the job its shape suggests.' Remind learners to wash hands if they handled any biological material and to carry microscopes with both hands.		
Explain Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: excretion homeostasis animals liver  READING: No closely matching reading in the ARES library for this activity. Search ARES for readings Search terms: excretion homeostasis animals liver	(25 min) Part 1 (10 min): Learners return to their predictions and revise them in a different colour, using evidence from the stations. They then build a short Concept Map/flowchart: Protein in a meal of beans or ugali with sukuma wiki, amino acids, liver (deamination), ammonia, urea, blood, kidney, urine. They add a second branch: old red blood cells, liver, bilirubin, bile, intestine, faeces, and a third branch: heat, sweat glands, sweat, evaporation, cooling. Part 2 (15 min): A resource person (clinical officer, nurse or laboratory technologist from the local health centre) gives a 10-minute talk on cirrhosis, hepatitis and fatty liver disease: causes (viruses, excess alcohol, fatty diet, obesity), symptoms (jaundice, swelling, tiredness) and prevention (hepatitis B vaccination, hygiene, balanced diet, avoiding alcohol). Learners prepare questions, then relate one disorder to a structural loss: for example, scarring in cirrhosis damages hepatocytes	Say: 'Pick up your coloured pen. Cross nothing out; show me how your thinking changed.' Draw the first branch of the concept map on the board and invite learners to complete the rest in their groups. Move between groups and ask: 'What links this arrow? What is the evidence?' WAIT TIME: 10-15 seconds before calling on anyone. Cold-call 5 learners to add one box or arrow each to a class map, asking each: 'What evidence from the stations supports that?' Introduce the resource person and set norms: 'We listen respectfully and ask scientific questions, not personal medical ones.' Before the talk, say: 'Write down one question you would like answered.' After the talk, ask 3 learners to give questions and discuss them. Pose: 'If cirrhosis scars the liver, which structure from our photomicrograph is affected and what happens to the waste in the blood?' WAIT TIME 10-15 seconds, then collect answers from 4 pairs. Correct	Concept Mapping/Flowcharts plus Think-Pair-Share. A flowchart forces learners to connect organs, substances and direction of movement, so liver, blood, kidney and skin become one system rather than separate facts. Applying the map to disorders makes learners reason from damaged structure to disrupted function, and to the animals in the phenomenon (panting dog, still lizard) to compare different strategies for heat loss.	Observable indicators: revised predictions show a correct statement that urea is made in the liver; concept maps include deamination, urea, bile and sweat with correctly directed arrows; at least 3 learners can explain why liver damage leads to jaundice. Concept check quiz (5 minutes, 4 questions): (1) Name the process that forms urea. (2) What pigment is excreted in bile? (3) Name two substances in sweat. (4) State one cause of fatty liver disease. Teacher scans for 70 percent correct and notes who needs re-teaching.

	and reduces blood flow so urea and bilirubin build up and jaundice results. They finish with a Think-Pair-Share: 'Which of our animals on the hot afternoon rely most on the skin to lose heat, and which rely little on it? Why?' (for example, humans and horses sweat, dogs pant because they have few sweat glands, lizards do not sweat).	misconceptions explicitly, for example: 'Urea is made in the liver and removed by the kidney.' Link back: 'So the goat's urea began in its liver. The kidney only cleared it.'		
Driving Question Board (DQB) Creation  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: excretion homeostasis animals liver  READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms: excretion homeostasis animals liver	(10 min) Learners add to the class Driving Question Board. Each group writes 1-2 new questions on sticky notes arising from today (for example: 'If sweat cools us, why do dogs pant instead?' 'Why do we feel thirsty after sweating?' 'How does the body know when to sweat?'). They also place sticky notes under 'Questions we can now answer' (for example, 'Where is urea made?') and 'Questions we still have'. The class sorts the questions into themes (water, temperature, sugar levels). Learners notice that many questions ask how the body keeps conditions constant when the outside changes, which leads to the next lesson.	Say: 'Look back at our big question: how do different animals maintain a balanced internal environment? What did the liver and skin add?' Give each group 2 minutes to write questions. WAIT TIME: 10-15 seconds after asking: 'What do you notice about these questions?' Cold-call 4 groups to read and place a note and explain the link to the phenomenon. Move answered questions to the 'Can answer' column with learners naming the evidence. Highlight three clusters and say: 'These questions are about keeping water, temperature and sugar steady. That is exactly what we investigate next lesson: how does the body keep internal water, temperature and sugar levels constant during external changes?' Photograph the DQB for records.	Driving Question Board (DQB) discussion. The DQB makes learners' questions visible, shows how evidence answers some of them while raising others, and gives learners ownership of the learning path. Sorting into themes turns scattered curiosity into a shared purpose for the next lesson.	Observable indicators: each group contributes at least one new, testable question; at least 2 notes are moved to 'Can answer' with a named piece of evidence; learners link a question to the phenomenon, for example 'dog pants because...'. Teacher records the question themes and which groups still ask only recall questions (what, where) instead of mechanism questions (how, why).
Model Building Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: excretion	(10 min) Learners revise the class model (a large poster of the whole body with the kidney, ureter, bladder, gills of fish, and Malpighian tubules from earlier lessons) by adding the liver and skin. In groups they add labelled arrows for: amino acids to liver to urea to blood to kidney; haemoglobin breakdown to bile to intestine; heat to sweat glands to	Say: 'Our model must now explain more of the phenomenon. Add the liver and skin, and show the direction of every arrow.' Provide markers and sticky notes so each group has a share of the poster. Circulate and ask: 'Where does the urea come from before the kidney gets it? What carries it?' WAIT TIME: 10-15 seconds. Cold-call 3 groups to explain one new addition	Model Building and Revision. Updating a shared model makes learners integrate new evidence with earlier evidence from the kidney, fish gills and insects, and shows what the model can and cannot yet explain (for example, why the dog pants and the lizard basks). Peer review builds scientific argumentation and shared responsibility.	Observable indicators: model includes the liver and skin with correct arrow directions and labels; exit tickets name deamination or sweating and give a correct function; at least 80 percent of exit tickets state one remaining gap (for example, the control of temperature or sugar). Peer review uses 'two stars and a wish' written on sticky notes. Teacher sorts exit

homeostasis animals liver  READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: excretion homeostasis animals liver	evaporation. Then they write an individual exit ticket explaining: 'Using the liver or skin, explain how a goat or a human removes wastes or loses heat on a hot afternoon in Kisumu, and what is still missing from our model?' Groups peer review each other's additions using 'two stars and a wish'.	to the class. Hand out exit tickets and say: 'Write in full sentences with at least two pieces of evidence.' Collect exit tickets at the door. Close with: 'We now know that more than one organ removes wastes, but we do not yet know how the body keeps water, temperature and sugar steady while the outside changes. That is our next question.'	tickets into 'secure', 'partly secure' and 'not yet' to plan a 5-minute reteach at the start of Lesson 5.
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D. TEACHER REFLECTION

1. Which structure-function link did learners find hardest (sinusoids, deamination or sweat glands), and what evidence in the annotated sketches and quiz showed this?
2. Did the stations give every learner a fair chance to see the images despite limited charts and slides? What would you change for large classes?
3. How well did the resource person link disease to the structure learners had just observed, and did learners ask scientific rather than personal questions?
4. Did learners still believe the kidney makes urea after the lesson? How do you know?
5. Which DQB questions show readiness for Lesson 5 on water, temperature and sugar regulation, and which learners need more support?
6. Were any learners uncomfortable discussing liver disease, and how did you handle it?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed photomicrographs and charts of the liver (hepatocytes, sinusoids, bile canaliculi) and skin (epidermis, dermis, sweat glands, capillaries), and heard about cirrhosis, hepatitis and fatty liver disease.
What did I learn?	The liver deaminates excess amino acids to make urea, breaks down haemoglobin to make bile pigments and processes toxic substances; the skin excretes sweat containing water, salts and a little urea and helps cool the body. Liver disorders damage these functions.
How does this explain the phenomenon?	Urea in the goat's urine begins in the liver, is carried by blood to the kidney and removed, while sweat or panting helps animals lose heat on a hot afternoon. The liver and skin work with the kidney to remove wastes and keep the body stable, but we still need to explain how water, temperature and sugar levels are kept constant.

LESSON 5 (80 minutes (double lesson)): Feedback Loops: How the Body Keeps Water and Blood Glucose Steady

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use feedback flowcharts and case studies to explain how negative feedback regulates water balance (ADH) and blood glucose (insulin and glucagon) in humans, and how this helps explain why goats and freshwater fish handle water so differently on a hot afternoon.
Knowledge	e) illustrate homeostatic regulation in humans. Learners explain that negative feedback detects a change from a set point and triggers a response that reverses it. They describe osmoregulation (osmoreceptors in the hypothalamus, ADH from the posterior pituitary, kidney collecting ducts) and blood sugar regulation (pancreatic islet cells, insulin, glucagon, liver). They distinguish diabetes mellitus from diabetes insipidus.
Skills	Constructing and interpreting flowcharts and concept maps (NGSS SEP: Developing and Using Models); analysing case study data (SEP: Analyzing and Interpreting Data); constructing evidence-based explanations (SEP: Constructing Explanations); asking clarifying questions (Critical Thinking and Problem Solving); collaboration in shared-resource groups.
Attitudes	Social Justice: shares resources equitably (charts, case study sheets, devices) within the group. Responsibility: takes on assigned roles and completes them. Appreciation of homeostasis as vital to health and of the need to conserve biodiversity (PCI: Environmental Conservation). Empathy for people living with diabetes.
Key Inquiry Question	1. How do animals get rid of waste products? 2. How do animals cope with environmental changes?
Purpose in Storyline	Lessons 1-4 established excretory products and organs and how the kidney forms urine. Lesson 5 shows that the kidney's output is not fixed but controlled by negative feedback. This explains why the goat (conserving water) produces concentrated urine while the freshwater fish produces dilute urine, and why the body's internal conditions stay steady. It also raises a new question for Lesson 6: how is core temperature kept stable when the weather changes, as in the panting dog and the basking lizard?
Safety Notes	Hand-washing after handling any specimens. For the glucose demonstration or sugary drinks, no tasting of lab materials. Handle case studies on diabetes sensitively; do not ask learners to disclose personal or family health conditions. If kidney specimens from earlier lessons are still on display, keep them covered, use gloves, and dispose of them as directed by the teacher. Ensure no learner is pressured to take part in any fasting or sugar activity.

B. LESSON OVERVIEW

Return to the anchoring phenomenon: the goat drinks water yet makes concentrated urine, while the freshwater fish continually excretes dilute urine. Learners first predict what the body does when it loses too much water on a hot day in Kisumu or Nairobi, and what happens to blood sugar after a meal of ugali and sukuma wiki. In groups of 5-6 (one set of charts and one device per group), they build feedback flowcharts for ADH osmoregulation and insulin/glucagon blood sugar control using homeostasis feedback charts. They then evaluate case studies of diabetes mellitus and diabetes insipidus to test their flowcharts by locating where each loop breaks. Using Predict-Observe-Explain, they apply the idea of nephron function from the kidney dissection to explain how ADH changes urine concentration. The class updates the Driving Question Board and revises its cumulative model by adding negative feedback control. The lesson ends with the question of how body temperature is regulated.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Hormone concentration metabolism and negative feedback <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Adh Effects On Blood Pressure <i>Related topic: not an exact match for this lesson.</i> Source: khan_academy  Search ARES for similar readings	Learners look at a short scenario on the board: Wanjiku runs 10 km in the hot afternoon at Kericho, sweats a lot and drinks little water. Later she eats a big plate of ugali and beans and a mandazi with sweet tea. In pairs (Think-Pair-Share) they individually sketch or write predictions for (a) what happens to her urine volume and concentration, and (b) what happens to her blood sugar after the meal and how the body brings it back to normal. They link back to the goat and the freshwater fish by predicting which animal's kidney is more like Wanjiku's after the run.	Display the scenario and the phenomenon picture (goat and fish). Say: "Take 2 minutes to think silently. Write what you think happens to her urine, and what happens to her blood sugar. Use arrows if it helps." WAIT TIME 10-15 seconds after each question before taking answers. Then: "Turn to your partner and share for 2 minutes. Look for where you disagree." Cold-call 4 pairs: "Pair 3, what did you predict happens to the urine? What made you think that?" Record predictions on the board in two columns (water, glucose) without correcting them. Ask: "Who predicted the body does something to reverse the change? What would that be?" Circulate and note common ideas such as 'the kidney just filters more'.	Think-Pair-Share surfaces prior ideas about the kidney and sugar and creates a record of initial thinking. Predicting before the evidence gives learners a stake in the investigation and lets them compare later ideas to their predictions, linking to Lessons 3-4 on urine formation.	Observable indicators: learners write a prediction with a reason, not just a statement; at least 3 of 4 cold-called pairs can state a cause-effect link (e.g., 'less water, so less urine'). Note on a checklist who mentions a control or reversing mechanism, and who holds misconceptions such as 'sugar is just removed in urine in everyone' or 'the kidney decides on its own'. Oral probes: "What is your evidence for that?"
Observe Phase  VIDEO: Hormone concentration metabolism and negative feedback <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Adh Effects On Blood Pressure <i>Related topic: not an exact match</i>	In groups of 5-6 with assigned roles (reader, flowchart builder, recorder, materials manager, reporter), learners use the homeostasis feedback charts to gather evidence. Part A: from the chart, they assemble an osmoregulation flowchart on manila paper using cards: dehydration (blood too concentrated) → osmoreceptors in hypothalamus → posterior pituitary releases more ADH → collecting duct becomes more permeable → more water reabsorbed → small volume of concentrated urine → blood water returns to normal → ADH falls. They then draw the opposite loop for drinking lots of water. Part B: they build a blood glucose flowchart: high glucose	Distribute materials: "Materials managers, collect one chart set, one case study sheet and the card set for your group. Share equally; every member must handle the cards." Model with the first card: "A negative feedback loop has a set point, a receptor, a control centre, an effector and a response. Find these in the chart." Circulate with a checklist, asking oral probes: "Which structure detects the change? Which one responds? Why do we call this negative feedback?" WAIT TIME 10-15 seconds after each probe. Cold-call 3 groups to explain one arrow: "Group 2, why does more ADH mean less urine?" Prompt for a link to Lesson 4: "Which part of the nephron does ADH act on?" For	Concept Mapping/Flowcharts: constructing the loops as cause-effect chains forces learners to identify receptors, control centres and effectors and to see why the change is reversed. Case studies act as a test of the model: learners use the flowchart to locate the failure that produces each set of symptoms. Collaborative role assignment builds equitable participation.	Observable indicators: flowchart shows a complete loop with a return arrow to the set point; ADH and insulin/glucagon are correctly placed; the case studies are correctly marked (diabetes mellitus: insulin missing or ineffective; diabetes insipidus: ADH missing or ineffective). Peer review: groups swap flowcharts and give one 'star' and one 'step' using a 3-point checklist (complete loop, correct hormones, correct direction of change). Oral probes recorded for at least 8 learners.

for this lesson. Source: khan_academy Search ARES for similar readings	after a meal → beta cells of islets of Langerhans release insulin → liver and muscle take up glucose and convert it to glycogen, cells use more → glucose falls; low glucose (after a long run or fasting) → alpha cells release glucagon → liver converts glycogen to glucose → glucose rises. Part C: they read two case studies. Case 1, diabetes mellitus (a 15-year-old in Nakuru with excessive thirst, frequent urination, glucose in urine, high blood glucose; Type 1 shows too little insulin). Case 2, diabetes insipidus (a patient in Kisumu passing up to 10 litres of very dilute urine daily, always thirsty, normal blood glucose, low ADH or kidney not responding to ADH). They mark on their flowcharts where the loop is broken in each case. Where possible, a group views the short feedback animation on the shared device (one device per group, rotating viewer roles).	case studies: "Underline the symptoms. Which step in your flowchart could fail to give these symptoms?" Extend with a question: "Why do diabetes mellitus and diabetes insipidus both cause thirst and a lot of urine, but only one has sugar in the urine?" Manage the large class by giving a 5-minute time check for each part, using group roles, and having groups with devices share with a neighbouring group if needed. Safety: remind about caring and respectful discussion of illness.		
Explain Phase  VIDEO: Hormone concentration metabolism and negative feedback <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Adh Effects On Blood Pressure <i>Related topic: not an exact match</i>	Learners return to their Predict-phase notes. Using Predict-Observe-Explain, each group writes a short explanation (4-6 sentences plus a labelled nephron sketch) of what happens in Wanjiku's body after the run and the meal, then explains how this reasoning applies to the goat and freshwater fish: the goat conserves water so ADH is high and urine is concentrated, while freshwater fish absorb water by osmosis and excrete dilute urine in large volume. Learners annotate their nephron diagrams from the kidney dissection to show where ADH acts (collecting duct) and where glucose is normally fully reabsorbed (proximal convoluted	Say: "Look back at your prediction. What did you get right, and what do you want to change? Use evidence from the flowchart." WAIT TIME 10-15 seconds. Cold-call 4 learners from different groups: "What would happen to Wanjiku's ADH level, and why?" and "What would happen to her blood sugar after the mandazi, and which hormone acts?" Press for reasoning with: "How do you know? Which part of the chart shows that?" Address misconceptions: ADH is not a waste product; insulin does not remove sugar from the body, it moves it into cells and stores it as glycogen; diabetes insipidus is unrelated to sugar. Challenge:	Predict-Observe-Explain closes the loop by having learners compare their predictions to the evidence and revise their reasoning. Writing a causal explanation with a diagram helps them connect the macro phenomenon (urine concentration, thirst) to the micro mechanism (hormones acting on the nephron).	Concept check quiz (5 items): 1) Define negative feedback. 2) Name the hormone released when blood is too concentrated. 3) State the effect of insulin on blood glucose. 4) Name the organ that converts glycogen to glucose under glucagon. 5) Give one difference between diabetes mellitus and diabetes insipidus. Success criterion: 4/5 correct. Annotated nephron diagram check: ADH correctly placed at the collecting duct, a correct arrow showing water reabsorption. Record learners below 3/5 for reteaching in the next lesson.

for this lesson. Source: khan_academy Search ARES for similar readings	tubule), explaining why glucose appears in the urine in diabetes mellitus. Then a concept check quiz (5 questions) is completed individually.	"Fish and goats both have kidneys. What does the idea of feedback tell us about why their urine differs?" Note that the fish also uses regulation of salt and water but not necessarily through the same hormone pathway, so learners should explain in terms of the direction of water movement. Administer the quiz: "Put pens down at the sound of the signal; no talking during the quiz." Collect for quick marking.		
Driving Question Board (DQB) Creation  VIDEO: Hormone concentration metabolism and negative feedback <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Adh Effects On Blood Pressure <i>Related topic: not an exact match for this lesson.</i> Source: khan_academy Search ARES for similar readings	Learners return to the Driving Question Board with the question: 'How do different animals maintain a balanced internal environment and eliminate toxic wastes despite drastic changes in their external conditions?' Each group writes on sticky notes (a) one question that has now been answered (with evidence from today) and (b) one new question. They place answered questions in an 'Answered' column with a short evidence note, and new questions in a 'Still to investigate' column. Likely new questions: 'How does the dog's panting help keep it cool?' 'Why does the lizard bask?' 'Do fish use ADH?' They also complete an exit ticket.	Say: "Look at the board. Which of our questions can we now answer? Bring your sticky note and explain the evidence." WAIT TIME 10-15 seconds. Cold-call 3 groups to place and justify notes: "What evidence from today supports your answer?" Guide the gap: "We have explained water and blood sugar. But look at the dog and the lizard. What have we not yet explained?" Draw out the temperature question and write it on the board as the lead-in: "How does the body keep core temperature stable when the outside weather changes drastically?" Hand out exit ticket slips and set 3 minutes: "Write one thing you understand, one thing you are unsure about, and draw the loop for one hormone."	The DQB makes progress visible and keeps the class focused on a shared, student-generated question. Sorting questions into answered and new builds a sense of purpose and shows how today's evidence connects to the next investigation.	Exit ticket: learners complete a mini-loop for either ADH or insulin/glucagon and state one question remaining. Success indicators: the loop includes stimulus, receptor, hormone, effector and response; questions refer to temperature or other animals. Quick sort of tickets into three piles: secure, partly secure, not yet secure, to plan the start of Lesson 6.
Model Building Phase  VIDEO: Hormone concentration	Groups revisit the class model of how animals maintain a balanced internal environment (started in earlier lessons and including excretory organs, kidney structure and urine formation). They add a	Say: "Take out your model from last lesson. What needs to be added or changed to show how the body keeps water and sugar steady?" WAIT TIME 10-15 seconds. Ask groups to revise for	Model Building & Revision: adding feedback loops shows that homeostasis is a dynamic process, not a static state. Peer review of models makes learners justify their representations with evidence and	Observable indicators for the model: includes labelled negative feedback loops with a return to the set point; correctly links ADH to the kidney and insulin/glucagon to the liver and pancreas; connects at

metabolism and negative feedback <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Adh Effects On Blood Pressure <i>Related topic: not an exact match for this lesson.</i> Source: khan_academy Search ARES for similar readings	layer for control: negative feedback loops with ADH and insulin/glucagon, placing arrows from the brain or pancreas to the kidney or liver. They add a link to the goat and the freshwater fish, showing high versus low ADH-type water conservation, and note what the model cannot yet explain (temperature, behavioural responses such as basking and panting). Groups do a gallery walk and peer review the physical or drawn models of two other groups, using sticky notes with one strength and one suggestion.	8 minutes. Prompt: "Where is the set point? What is the feedback arrow? What happens if that part fails?" During the gallery walk say: "Visit two other groups. Give one strength and one suggestion; use evidence, not opinion." Cold-call 3 groups to share a change they made and why: "What did you change, and what evidence made you change it?" Close by linking to the next lesson: "Our model explains water and sugar. It cannot yet explain why the dog pants or the lizard basks. In Lesson 6 we will investigate temperature regulation." Ask learners to store models safely for later revision.	identify gaps that motivate the next lesson.	least one loop to the goat or the fish; records one limitation. Peer review sticky notes use evidence-based comments. Oral probe: "What would happen to the model if ADH were missing?" Record how many groups can answer with diabetes insipidus symptoms.
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D. TEACHER REFLECTION

Which learners could describe the feedback loop in their own words and which only copied the chart? Did learners confuse diabetes mellitus and insipidus, and which case-study prompts cleared this up? Was the group size right for sharing the charts and the single device? Did the quiz and exit tickets show who needs reteaching on hormones and nephron locations? How well did learners link the loops back to the goat, the fish, and the other animals? Which new questions on the DQB lead naturally into temperature regulation for Lesson 6?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that hormone signals such as ADH, insulin and glucagon form feedback loops, and that in diabetes mellitus and diabetes insipidus the symptoms (thirst, lots of urine, sugar in urine or very dilute urine) show where a loop has failed.
What did I learn?	We learned that negative feedback detects a change from a set point and triggers a response that reverses it; ADH adjusts how much water the kidney reabsorbs, and insulin and glucagon keep blood glucose within a narrow range.
How does this explain the phenomenon?	We can explain why the goat can produce concentrated urine to save water while the freshwater fish excretes dilute urine, how the body keeps water and glucose steady after exercise and a meal, and why the next question is how body temperature is kept stable in animals like the dog and the lizard.

LESSON 6 (80 minutes (2 x 40-minute periods)): Keeping Cool and Staying Warm: Thermoregulation in the Dog, the Lizard and Us

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use thermoregulation diagrams, video clips and field observation notes of local animals to figure out how physiological mechanisms (sweating, vasodilation, vasoconstriction) and behavioural adjustments (basking, panting, burrowing, hibernation, aestivation) control body heat, and use feedback loops to explain why the dog pants and the lizard stays still on a warm rock.
Knowledge	Learners will explain that thermoregulation is a homeostatic process that keeps body temperature within a narrow range through negative feedback. They will describe how the skin and hypothalamus respond to heat (vasodilation, sweating, panting) and to cold (vasoconstriction, shivering, hair erection). They will distinguish endotherms such as dogs, goats and humans from ectotherms such as lizards, which rely on behaviour (basking, shade-seeking, burrowing). They will also explain hibernation and aestivation as responses to prolonged cold or heat and drought.
Skills	Developing and using models (flowcharts of feedback loops); analysing and interpreting data (field observation notes and temperature data); constructing explanations from evidence; asking questions (Critical Thinking and Problem Solving); using digital devices in groups to view video clips (Digital Literacy).
Attitudes	Responsibility in carrying out assigned group roles during observation; Social Justice in sharing devices and materials equitably; appreciation of local biodiversity and its conservation (Environmental Conservation PCI); curiosity about how animals in the community survive environmental change.
Key Inquiry Question	How do animals cope with environmental changes? (KICD Key Inquiry Question 2)
Purpose in Storyline	Lessons 1 to 5 built evidence on excretory structures and products, the kidney and nephron, urine formation, and the liver and skin. Lesson 6 turns to the other half of the driving question: keeping a steady internal environment as external conditions change. It explains the dog and the lizard in the anchoring phenomenon and shows that temperature control uses the same sensor, control centre, effector and negative-feedback logic as osmoregulation. This raises the question for Lesson 7: Why are excretion and homeostasis so vital for the overall survival of living organisms?
Safety Notes	Conduct field observations from a safe distance. Do not handle lizards, snakes or unfamiliar dogs and goats, and watch out for ticks and sun exposure. Learners wear hats and carry drinking water outdoors. If the class stays indoors, use video clips and observation notes instead. Do not place hot-water or ice demonstrations on bare skin; use lukewarm and cool water only. Wash hands after any outdoor activity. Share devices in groups of 5 to 6 and keep volume low.

B. LESSON OVERVIEW

Learners return to the hot afternoon scene: a dog panting, a lizard motionless on a warm rock, goats drinking and fish excreting dilute urine. They first predict why the dog pants and the lizard stays still. In the Observe phase, groups of 5 to 6 share one device or printed set to study thermoregulation diagrams, video clips and field observation notes of local animals (dogs, goats, agama lizards, tortoises, lungfish, frogs, cattle seeking shade at midday in the Rift Valley) responding to weather. They trace feedback loops for heat and cold in a mammal and compare these with the behavioural strategies of an ectotherm. In the Explain phase they build flowcharts and write evidence-based explanations of the dog and the lizard. They then add questions and evidence to the Driving Question Board and revise their cumulative model by adding a temperature-control loop alongside the osmoregulation and excretion components. An exit ticket leads to the question for Lesson 7 on why excretion and homeostasis are vital for survival.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Hormone concentration metabolism and negative feedback <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Thermoregulation Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners recall the anchoring scene: the dog panting on a hot afternoon in the community and the lizard sitting motionless on a warm rock. Individually, they write a prediction for 2 minutes answering: 'Why does the dog pant while the lizard stays still on a hot rock? What is each animal doing to its body temperature?' They then do Think-Pair-Share, sketch a quick 'before' arrow diagram of how heat leaves or enters each animal, and record a confidence rating from 1 to 3.	Display or sketch the scene on the board. Say: 'Picture Kericho at 2 pm. A dog is panting, a lizard is on a hot rock. Write down what each animal is doing to its body temperature, and why.' Give 2 minutes of silent writing, then WAIT TIME of 10 to 15 seconds before pairing. Say: 'Turn to your partner and compare for 1 minute; look for where you disagree.' Cold-call 4 pairs from different parts of the room, including at least 2 who have not spoken this term, and record competing ideas on the board without judging, such as 'dogs sweat like us' or 'lizards are lazy'. Probe: 'What makes you think that? What evidence would change your mind?' Mark unresolved ideas with a star as questions for the Observe phase.	Think-Pair-Share (Predict phase of Predict-Observe-Explain): writing a prediction first surfaces learners' prior ideas (for example, that all animals sweat, or that stillness means laziness) so the later evidence can confront them. Pair talk lets every learner rehearse reasoning before the whole class hears it, which supports equitable participation in a large class.	Oral probes and prediction sheets. Observable indicators: learners mention heat loss or gain, body temperature, or a link to the environment; learners distinguish panting from sweating; learners give a reason for the lizard's stillness. Note how many predictions use words such as 'cool down', 'warm up' or 'body temperature' versus only describing behaviour. Tally misconceptions on the board to guide the Observe phase.
Observe Phase  VIDEO: Hormone concentration metabolism and negative feedback <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Thermoregulation Source: Khan Academy (English	In groups of 5 to 6, with assigned roles (reader, recorder, diagram analyst, device operator, reporter) that rotate between stations, learners work through 3 stations in about 25 minutes. Station 1, Diagrams: they study a thermoregulation diagram of a mammal's skin and hypothalamus on a printed chart, tracing what happens when the blood is too hot or too cold (vasodilation, sweating, panting, vasoconstriction, shivering, hair raising), and annotate the arrows. Station 2, Video clips on the shared device: they watch short clips of a dog panting, an agama lizard basking, shade-seeking and moving into a	Before starting, say: 'Your job is to collect evidence of how the body temperature of these animals is controlled. Record what you SEE, not what you assume.' State the safety rules and the roles, and set a timer for 8 minutes per station. Circulate and ask each group at least one probing question: 'What is the blood vessel in the skin doing here, and what does that do to heat loss?' 'Why might the lizard move into shade at 2 pm?' 'How is the dog's panting different from the lizard's behaviour?' After each question WAIT 10 to 15 seconds. If a group says 'the lizard is lazy', respond: 'What does the evidence in your notes show about when it	Evidence gathering through multiple representations (diagram, video, field notes): comparing the diagram of a mammal's feedback system with real behaviours of local ectotherms and endotherms helps learners see that organisms keep internal conditions steady by different means. Structured recording tables separate observation from inference, and rotating roles ensure shared responsibility.	Annotated thermoregulation diagrams and observation tables. Observable indicators: the arrows correctly show vasodilation bringing blood close to the skin surface to lose heat and vasoconstriction reducing blood flow to the skin to conserve heat; the table lists at least 4 animal behaviours with matched environmental conditions; learners classify each as physiological or behavioural. Use a 3-question concept check quiz by show of cards (A/B/C) at the end: for example, 'Which response lowers body temperature: vasoconstriction, vasodilation or shivering?' Record the number of

- US curriculum) Search ARES for similar readings	rock crevice, cattle or goats resting under acacia trees at midday, and, where available, a tortoise or lungfish aestivating or a hibernating animal. They record what the animal does, the environmental condition, and the likely effect on body temperature. Station 3, Field observation notes: they read notes collected by the teacher or learners, for example 'Agama lizard on rock at 9 am, 11 am and 2 pm; moves into shade at 2 pm', 'Dog under mango tree panting at midday', 'Goats rest in shade at noon, graze at 5 pm', and 'Frogs buried in mud during the dry season at Lake Victoria shores'. They tabulate time of day, behaviour, temperature trend and the type of mechanism (physiological or behavioural). Learners who cannot access the device view the clips with another group in turns so that everyone gets a fair turn.	moves?' Where devices are limited, rotate one device across groups and give the others the printed notes first. Pause for 2 minutes at the end and ask 3 cold-called learners to name one observation that surprised them.		correct answers to decide which ideas to revisit.
Explain Phase  VIDEO: Hormone concentration metabolism and negative feedback <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Thermoregulation Source: Khan Academy (English - US curriculum) Search ARES for similar	Groups build a concept map or flowchart of a negative feedback loop for body temperature, labelling stimulus (rise in body temperature), receptors (thermoreceptors in skin and hypothalamus), control centre (hypothalamus), effectors (sweat glands, skin arterioles, muscles) and response (sweating, vasodilation, panting) and then the reverse loop for cold (vasoconstriction, shivering, hair erection). They then add a branch for behavioural responses. Using their evidence, each group writes a 3 to 4 sentence explanation for the dog and the lizard: the dog (an endotherm) loses heat through panting, which evaporates water from the tongue and airways, and	Say: 'Use your evidence to show how the loop works. Every arrow must be justified by something you observed.' Give 10 minutes to build the flowchart and 5 minutes to write the explanations. Circulate and ask: 'What happens next if the body is still too hot? How does the loop stop?' 'Why is this called negative feedback?' WAIT 10 to 15 seconds. Then run a gallery walk: groups swap charts and leave 1 sticky note with a question or suggestion. Cold-call 3 groups to explain the dog and 3 groups to explain the lizard, and ask the class to compare: 'What is the same and what is different?' Address any misconception that the dog sweats through its tongue, or that the lizard produces its own	Concept mapping/flowcharts of homeostatic feedback loops: laying out stimulus, receptor, control centre, effector and response turns separate facts into a causal chain and makes the idea of negative feedback visible. Peer review in the gallery walk helps learners test their reasoning against others and refine their explanations of the anchoring phenomenon.	Peer-reviewed flowcharts and written explanations. Observable indicators: all 5 loop components are present and ordered correctly; the cold loop shows vasoconstriction and shivering; the explanation names panting or vasodilation for the dog and basking, shade-seeking or burrowing for the lizard; learners note that ectotherms depend on the environment. Use a rubric of 'Complete / Partly complete / Not yet' for each criterion. Oral probes record whether learners can state why the loop is 'negative'.

readings	through vasodilation; the lizard (an ectotherm) cannot regulate heat internally and uses behaviour, so it basks to gain heat, then moves into shade or a crevice when too hot. Groups also match other local examples (burrowing by mole rats and termites, hibernation, aestivation in lungfish and some frogs and snails during dry spells in northern Kenya) to the problem each solves. Finally, learners link back to the goats and fish: water balance and temperature control both use feedback loops.	heat. Prompt: 'Why does panting also cause the dog to lose water? How might that link to what we learned about the goats and kidney function?'		
Driving Question Board (DQB) Creation  VIDEO: Hormone concentration metabolism and negative feedback <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Thermoregulation Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners revisit the class Driving Question Board and the question 'How do different animals maintain a balanced internal environment and eliminate toxic wastes despite drastic changes in their external conditions?' Each group writes one answered question (for example, 'Why does the dog pant?') on a sticky note with a short evidence summary and moves it to the 'Answered' section. Each group also writes one new question (for example, 'What happens to the kidney and the body if an animal loses too much water through panting or sweating?' or 'Why do excretion and homeostasis matter for survival?') and adds it to the board. The class reads the new questions and sorts them into clusters such as water balance, temperature and survival.	Say: 'Look at our board. Which of our questions can we now answer, and what evidence do we have?' Give 3 minutes for groups to write, then WAIT 10 to 15 seconds as notes are placed. Cold-call 3 groups to read an answered question and justify it with evidence from today's stations. Ask: 'Which new questions connect temperature control with what we learned about the kidney and sweat?' Highlight the question about why excretion and homeostasis are vital for the overall survival of organisms and say: 'This will be our focus next lesson.' Praise questions that ask 'how' and 'why' rather than 'what'.	Driving Question Board discussion: displaying answered and new questions makes learning progress visible and keeps learners' own questions steering the unit. Clustering links today's evidence to earlier lessons on urine formation and the skin, and gives purpose to the next investigation.	Quality of sticky-note questions and evidence statements. Observable indicators: answered notes cite specific evidence (a diagram or field note); new questions link two ideas, such as water loss and kidney function; at least half the groups contribute a question that points to survival. The teacher notes patterns of misunderstanding to address in Lesson 7.
Model Building Phase  VIDEO: Hormone concentration metabolism and	Learners revise their cumulative class or group model of how animals maintain a balanced internal environment and remove wastes. Using poster paper, they keep the earlier components (excretory organs, nephron and	Say: 'Add today's evidence to your model. Use a new colour for everything you add or change, and show how thermoregulation connects to water balance and waste removal.' Give 12 minutes. Circulate and ask: 'Where does the	Model building and revision: adding thermoregulation to the evolving model lets learners integrate temperature control with excretion and osmoregulation into one coherent explanation of homeostasis. Colour-coded	Peer review of models and exit tickets. Observable indicators: the model shows a labelled feedback loop with sensor, control centre and effector; it separates physiological from behavioural responses; the dog and lizard are

negative feedback <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Thermoregulation Source: Khan Academy (English - US curriculum) Search ARES for similar readings	urine formation, liver and skin) and add a new thermoregulation section with a feedback loop and a panel for behavioural strategies. They use colours for new additions, label the dog and lizard, and add the goats and fish to show osmoregulation. In pairs, they swap models and give peer review using 2 stars (what is well represented) and a wish (what is missing or unclear). Finally, learners complete an exit ticket: (1) Draw or write the steps of the loop that cools a hot body; (2) Explain in two sentences why the lizard stays still on the hot rock while the dog pants; (3) Write one question about why excretion and homeostasis are vital for survival.	heat go when the dog pants? What does that cost the dog in water?' 'How would your model show the goat in the same heat?' WAIT 10 to 15 seconds after each question. Facilitate peer review with the prompt: 'Give two stars and a wish.' Cold-call 3 pairs to share one revision they made and why. Collect exit tickets at the door. Close by saying: 'If every system works to keep the body steady, what could happen when it fails? That is our question for next lesson.'	revisions make conceptual change visible, and peer review builds a shared standard for a good model.	correctly explained; connections to water balance appear. On exit tickets, a successful response correctly orders the cooling steps (temperature rise detected, hypothalamus signals, vasodilation and sweating or panting, temperature falls) and contrasts endotherm with ectotherm. Sort tickets into 'secure', 'developing' and 'needs support' to plan reteaching.
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D. TEACHER REFLECTION

Before the lesson, did I collect or prepare field notes, video clips and printed diagrams so that groups sharing one device still get equal access? After the lesson, which learners could explain the loop in their own words and which only copied labels? Did any hold the idea that the lizard is 'lazy' or that dogs sweat through their tongues, and did the evidence change their thinking? Were the cold-calls spread across the whole room? How well did learners connect temperature control to water balance and kidney function, and which students need support with negative feedback? What adjustments are needed to the DQB clusters to prepare for Lesson 7 on why excretion and homeostasis are vital for survival?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed a dog panting and cattle and goats resting in shade at midday, an agama lizard basking on a rock in the morning and moving into shade or a crevice at midday, and, in video clips and notes, animals that burrow, hibernate or aestivate during extreme cold or dry conditions. We also saw diagrams of blood vessels in the skin widening (vasodilation) and narrowing (vasoconstriction) and of sweating and shivering.
What did I learn?	Body temperature is kept within a narrow range by negative feedback: thermoreceptors detect a change, the hypothalamus acts as control centre, and effectors such as sweat glands, skin arterioles and muscles reverse the change. Endotherms like dogs, goats and humans use physiological responses (sweating, panting, vasodilation, vasoconstriction, shivering) plus behaviour, while ectotherms like lizards rely mainly on behaviour (basking, shade-seeking, burrowing). Hibernation and aestivation help animals survive prolonged cold or hot, dry periods.
How does this explain the phenomenon?	The dog pants because evaporation of water from its tongue and airways carries heat away, and its skin blood vessels dilate to lose heat, keeping its body temperature steady. The lizard stays still on the warm rock because it cannot control its temperature internally, so it gains heat by basking and moves into shade when it gets too hot. Both are using homeostasis to survive heat,

	which raises the question of why excretion and homeostasis are so vital for the survival of all living organisms.
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LESSON 7 (2 lessons of 40 minutes each (80 minutes)): Putting It All Together: Final Explanation of How Animals Keep a Balanced Internal Environment and Remove Toxic Wastes

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners synthesise evidence from Lessons 1-6 into a final explanatory model that accounts for the anchoring phenomenon (panting dog, basking lizard, goats producing concentrated urine, freshwater fish producing dilute urine). They compare it with their Lesson 1 model and present answers to the Driving Question Board (DQB).
Knowledge	By the end of the lesson, the learner should be able to: (a) relate excretory structures and their products in insects, birds, freshwater and marine fish, and mammals; (b) relate the internal structure of the mammalian kidney (cortex, medulla, pelvis, nephron) to its function in urine formation; (c) relate the structure of the liver and skin to their excretory functions; (d) illustrate homeostatic regulation in humans (osmoregulation, thermoregulation, blood sugar regulation) using negative feedback; (e) reflect on the importance of excretion and homeostasis in animals.
Skills	Developing and using models (revising a cumulative model); Constructing explanations (claim-evidence-reasoning); Engaging in argument from evidence (peer review of models); Analysing and interpreting data (comparing Lesson 1 and final models); Obtaining, evaluating and communicating information (presenting DQB answers); Asking questions to be clear on what is needed (Critical Thinking and Problem Solving).
Attitudes	Responsibility: engaging in assigned roles and duties during group work and presentations. Social Justice: sharing resources (devices, specimens, charts) equitably. Environmental Conservation and Waste Management: appreciating the need to conserve biodiversity by recognising how local animals respond to environmental changes. Observing laboratory safety when handling kidney specimens.
Key Inquiry Question	1. How do animals get rid of waste products? 2. How do animals cope with environmental changes? Driving Question: How do different animals maintain a balanced internal environment and eliminate toxic wastes despite drastic changes in their external conditions?
Purpose in Storyline	This is the final lesson of the unit. Learners have gathered evidence on excretion in insects, birds and fish, kidney structure and dissection, urine formation, liver and skin, human homeostasis (osmoregulation, thermoregulation, blood sugar regulation, diabetes mellitus and insipidus), liver disorders and animal behavioural responses (basking, panting, hibernation, burrowing, aestivation). Now they unify it into one explanation of the phenomenon. Core idea: keeping a constant internal environment (homeostasis) and clearing metabolic waste (excretion) are essential for cellular survival and overall health. This is a unit reflection, so no new question is raised.
Safety Notes	Any kidney specimens revisited for the Predict-Observe-Explain check must be fresh goat or cow kidney from a local butcher, handled with gloves, on trays, with scalpels used only by the assigned group member. Wash hands with soap and water after handling. Dispose of specimens by burying in a pit or as the school directs. Keep food and drink away from specimens. Limit digital devices to group viewing, with cables and chargers kept off walkways.

B. LESSON OVERVIEW

Learners return to the hot-afternoon scene: a panting dog, a still lizard on a warm rock, goats making concentrated urine and freshwater fish making large volumes of dilute urine. Over two periods they (1) predict, then check, how a kidney and feedback loop explain these differences, (2) build a final model with claim-evidence-reasoning statements, (3) compare it with their Lesson 1 model, (4) close the DQB by presenting answers to every question, and (5) reflect on why excretion and homeostasis matter for cell survival and health. Large classes work in groups of 5-6 with one shared device and one kidney or diagram set per group. Teacher walks the room with the rubric sheet.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: The kidney and nephron Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Mechanisms of transport: tonicity and osmoregulation Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners look at the phenomenon image or the teacher's re-telling of the scene (dog, lizard, goat, tilapia from Lake Victoria). Individually they write a 2-minute prediction: 'If a goat is deprived of water for a day on a hot afternoon in Kajiado, what will happen to (a) its urine concentration, (b) its body temperature, (c) its blood water level, and which organ and hormone are involved?' They then Think-Pair-Share with a partner and add one sentence connecting their prediction to the lizard or the fish.	Display the phenomenon image and say: 'Remember the dog, the lizard, the goat and the fish?' Today you will explain them using everything we have learned.' Read the prompt aloud. Give WAIT TIME of 10-15 seconds of silent thinking before writing. Circulate and note 3 predictions to share. Say: 'Turn to your partner and explain WHY, not just WHAT.' Cold-call 4 pairs, asking: 'What did your partner say that you had not thought of?' Record the predictions on the board under 'Predictions to test'. Do not correct yet.	Think-Pair-Share combined with the Predict step of Predict-Observe-Explain. Committing to a prediction makes learners aware of their current ideas, so that the later evidence check shows how their thinking has changed. Pairing exposes alternative ideas such as 'goats just drink less' versus 'the kidney conserves water through ADH'.	Oral probes and prediction sheets. Indicators: learner mentions kidney or nephron and water reabsorption for the goat; links high temperature to sweating, panting or behaviour; uses the word 'balance' or 'constant'; notes common misconceptions (e.g. 'fish do not need to excrete because they live in water'). Teacher tallies how many of the 4 cold-called pairs mention a hormone or feedback.
Observe Phase  VIDEO: The kidney and nephron Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Mechanisms of transport: tonicity and osmoregulation Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Groups of 5-6 review their evidence set: (1) the labelled kidney and nephron drawings from the dissection, (2) the urine-formation animation, watched again in groups on one shared device for 3 minutes, (3) the feedback-loop charts for thermoregulation, osmoregulation and blood sugar, and (4) the comparison table of nitrogenous wastes (ammonia in freshwater fish, urea in mammals, uric acid in insects and birds). Using a fresh goat or cow kidney from the local butchery, if available, each group does a quick Predict-Observe-Explain check: predict where the cortex, medulla and pelvis will be on a cut surface, locate them, and explain how each relates to filtration, concentration of urine and collection. Each group fills the	Assign roles (specimen handler, recorder, device operator, reporter, safety monitor) and say: 'Everyone has a role, and roles rotate so that resources are shared fairly.' Review the safety rules before gloves go on. Give 2 minutes of WAIT TIME for groups to predict the layers of the kidney before they observe. Circulate with the rubric and ask: 'Which part of this kidney makes sure the goat loses little water?' and 'Why does the fish lose so much water, and which part of its body is losing it?' Cold-call 3 reporters to state one observation each. Pause the animation at the loop of Henle and ask: 'What is happening to water here?' Wait 10 seconds before taking answers.	Predict-Observe-Explain on the kidney specimen, plus an evidence table. Learners test their mental model against real tissue and tie structure to function. The table forces them to compare across animals (insects, birds, fish, mammals) so patterns emerge: the waste product and the strategy depend on the water available and the energy cost.	Annotated kidney/nephron diagrams and oral probes. Indicators: correct labelling of cortex, medulla, pelvis, glomerulus, Bowman's capsule, loop of Henle and collecting duct; correctly states that the glomerulus filters and the tubule reabsorbs; evidence table has at least 4 animals correctly matched to waste product and strategy; safe handling observed (gloves, trays, hand washing). Teacher ticks a checklist for each group.

	evidence table: Animal Environmental challenge Waste product Organ or behaviour How it keeps balance.			
Explain Phase  VIDEO: The kidney and nephron - dubbed(KY) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (Kiswahili)  Search ARES for similar videos  EXERCISE: Mechanisms of transport: tonicity and osmoregulation Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Each group writes a Claim-Evidence-Reasoning (CER) explanation for the four animals in the phenomenon. Example claim: 'Different animals keep a stable internal environment using structures and behaviours suited to their habitat.' Evidence is cited from the lessons (e.g. panting increases evaporation to cool the dog; basking lets the ectothermic lizard raise its body temperature, and it moves to shade when too hot; the goat's long loops of Henle and ADH produce concentrated urine; the freshwater fish gains water by osmosis and excretes large volumes of dilute urine). Groups then draw a concept map or flowchart of a negative feedback loop (stimulus → receptor → control centre → effector → response → return to set point) for one example: thermoregulation in the dog, blood sugar regulation (insulin/glucagon, linking to diabetes mellitus) or osmoregulation (ADH, linking to diabetes insipidus). They add the liver (deamination, urea formation) and skin (sweat) to show where wastes come from and leave.	Say: 'Your explanation must answer WHY the dog pants, WHY the lizard stays still, and WHY the goat and the fish differ.' Provide the CER frame on the board. Allow 10-15 seconds of WAIT TIME after each question before calling on anyone. Prompt: 'What would happen to the loop if the effector stopped working? Which disorder have we studied that fits?' Cold-call 5 learners from different groups to share one link in their feedback loop and ask each: 'What is your evidence?' Invite a different group to challenge: 'Do you agree? Why or why not?' Address misconceptions, e.g. that lizards 'do not control' temperature (they do so behaviourally), or that panting 'makes the dog lose waste' (it mainly loses heat by evaporation).	Concept Mapping/Flowcharts of homeostatic feedback loops, plus CER. Drawing the loop makes causal chains visible and shows that homeostasis in all examples follows the same logic, while CER links claims to evidence. This moves learners from lists of facts to mechanisms.	Concept check quiz (5 questions, 5 minutes, done individually) and oral probes. Indicators: CER has a claim, at least two pieces of evidence and reasoning that mentions negative feedback; flowchart has all six components in correct order and the arrows show reversal back toward the set point; learner correctly distinguishes ectotherm behaviour from endotherm physiology; at least 4 of 5 quiz items are correct. Teacher records learners scoring below 3 for follow-up.
Driving Question Board (DQB) Creation  VIDEO: Hormone concentration metabolism and negative feedback	Learners go to the class DQB wall. Each group takes 3-4 sticky notes of questions from earlier lessons (e.g. 'Why does the lizard bask?', 'Why do fish urinate so much?', 'What happens if the liver fails?', 'Why do diabetics pass a lot of urine?'). They write an answer on a new note with evidence, place it next to the question, and mark	Say: 'Our DQB is not finished until each question has an answer backed by evidence.' Give each group a colour of marker. Before presentations, allow 10-15 seconds of WAIT TIME for groups to rehearse silently. During presentations ask the audience: 'What evidence supports that answer?' Cold-call 3 learners from	Driving Question Board discussion. Revisiting learner-generated questions shows that the unit was driven by their curiosity, and presenting evidence-based answers consolidates learning through explanation. Marking questions Answered/Partly/New builds metacognition about what they	Peer review and oral probes using the rubric. Indicators: each answer cites at least one piece of evidence from the unit; presenters use correct terms (osmoregulation, reabsorption, negative feedback, deamination); audience members accurately paraphrase; at least 80% of DQB questions are marked Answered or Partly answered with

<i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Mechanisms of transport: tonicity and osmoregulation Source: Khan Academy (English - US curriculum) Search ARES for similar readings	each question as Answered, Partly answered or New. Groups present 2 answers to the class in 90 seconds each, using their model. The class identifies remaining questions as areas for further study (such as local liver disease and community health or conservation of local wildlife during drought), without launching a new unit.	other groups to paraphrase an answer. Use the rubric sheet to score presentations. Point out that answers link to both KICD key inquiry questions: waste removal and coping with environmental change. Close with: 'Which question taught you the most?' and take 3 volunteers.	know.	justification. Teacher notes unresolved misconceptions.
Model Building Phase  VIDEO: The kidney and nephron Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Mechanisms of transport: tonicity and osmoregulation Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Groups take out the Lesson 1 initial model (their first drawing of how the dog, lizard, goat and fish cope). On a large sheet they draw the Final Model showing: (1) external changes (heat, water availability, salinity), (2) sensors and control (brain, hormones such as ADH, insulin), (3) organs for excretion (kidney, liver, skin, gills, Malpighian tubules), (4) wastes (ammonia, urea, uric acid, CO₂), (5) behaviours (basking, panting, burrowing, hibernation, aestivation), and (6) the outcome: stable internal conditions that keep cells alive. They use a different colour for the changes from Lesson 1 and write 'I used to think... Now I think... because...'. Groups swap models for a gallery walk and peer review with two stars and a wish. Finally, individually, learners complete the exit ticket and the summary table (below), reflecting on why excretion and homeostasis matter, for example how kidney failure or liver disease affects a person's	Say: 'Your final model must explain all four animals, not just one.' Display success criteria: includes cause-and-effect arrows, all three wastes, one feedback loop, and at least two behaviours. Give 10-15 seconds of WAIT TIME before groups begin drawing so they plan. Circulate and ask: 'Where on your model is the goat's water saved?' and 'Where does the fish lose water balance, and how does it fix it?' During the gallery walk, stand at the exit and cold-call 4 learners: 'Name one change you made from your first model and the evidence that made you change it.' Collect exit tickets. Conclude: 'Excretion and homeostasis keep every cell alive; when they fail, health suffers.' Do not introduce a new question; this is the unit reflection.	Model Building and Revision with Lesson 1 versus final comparison. Putting earlier and final models side by side makes conceptual change visible and shows the learner how evidence has improved their explanation. Peer review through the gallery walk develops argumentation skills.	Peer review of physical models, exit tickets and the rubric. Indicators: final model includes all six elements and explains each of the four animals; at least three specific revisions from Lesson 1 are labelled with reasons; peer feedback is specific (two stars and a wish); exit ticket states one way excretion protects cells and one way homeostasis protects health, using correct vocabulary. Proficiency: 4 of 5 rubric criteria met.

	health in their community.			
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D. TEACHER REFLECTION

After the lesson, consider: Did learners explain all four animals with the same underlying principles (balance of water, temperature and wastes via feedback), or only recall isolated facts? Which groups still confuse ectothermic behaviour with endothermic physiology, or ammonia/urea/uric acid with habitat? Were devices and kidney specimens shared fairly in large groups, and did every learner have a real role? Which DQB questions remain Partly answered and why? Compare Lesson 1 and final models for 5 learners to judge growth, and plan remedial support (for example, a short revision on the nephron) for those who scored below the proficiency threshold. Note any safety lapses during the specimen check and how the next class can avoid them.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	What we observed: the panting dog, the basking lizard, goats producing concentrated urine and freshwater fish producing large volumes of dilute urine; the layers of the goat or cow kidney (cortex, medulla, pelvis); wastes in insects, birds, fish and mammals; and how feedback loops work.
What did I learn?	What we learned: homeostasis keeps temperature, water, salts and blood sugar steady using negative feedback; excretion removes toxic metabolic wastes such as ammonia, urea and uric acid; the kidney, liver and skin do this work in mammals; and behaviours such as basking, panting, burrowing, hibernation and aestivation help animals cope with environmental change.
How does this explain the phenomenon?	Our final explanation: Animals face different environments, but all must keep their cells in a stable internal environment and remove toxic wastes. The dog pants to lose heat; the lizard basks or shelters to adjust its temperature behaviourally; the goat's nephrons reabsorb water to make concentrated urine; the freshwater fish gains water by osmosis and excretes dilute urine. Because each structure or behaviour suits the animal's habitat, its cells survive and it stays healthy. This is why excretion and homeostasis are essential.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.